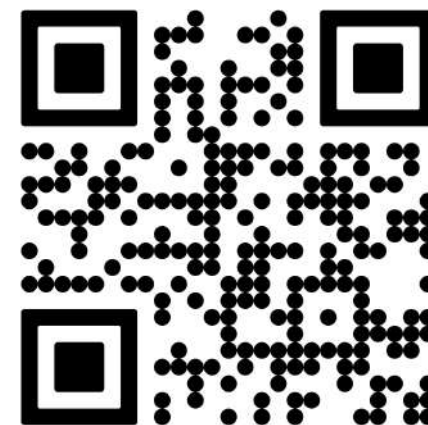


SCG YOUNG LEADER

Abridged Business Concept 2026

Accelerate critical capabilities building through workshop with case practices
โปรแกรมการพัฒนาที่สำคัญผ่านการจัดการเรียนรู้พร้อมกรณีฝึกปฏิบัติ



SCG ABC 2026

Agenda	Day 1	Day 2	Day 3	Day 4
	08.30-12.00	08.30-12.00	08.30-12.00	08.30-12.00
	Ice Breaking Activity Cross Silo Collaboration Trust & Empowerment	Business Acumen • Business Value Chain • Ecosystem & Partnership • Green Innovation & Inclusive Society	Digital Dexterity & Data Intelligence • Strategic Technology • Generative AI • Data for Impact	Marketing & Commercial • Marketing & Sales Strategy • AI-Powered Marketing & Sales • Social & Live Commerce
	13.00-16.30	13.00-16.30	13.00-16.30	13.00-16.30
	People & Organization • Self-Awareness & Inspiring • DEI • Agile Leadership & Green Innovation	Innovative Entrepreneurship • Business Model Navigator • Opportunity Scouting & Concept • Venture Building	Supply Chain & OpEx • Supply Chain Transformation • Next-Gen OpEx • Sustainable Value Chain & ESG	Bring it Altogether Case Practice

Digital Dexterity & Data Intelligence

Prof. Peerapon Vateekul, Ph.D.

Department of Computer Engineering, Faculty of Engineering
Chulalongkorn University

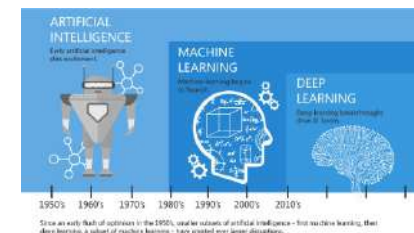
Peerapon.v@chula.ac.th

<https://github.com/pvateekul/iris-scg2026v2>



Prof. Peerapon Vateekul, Ph.D. (Aj.Pop)

- Professor at Department of Computer Engineering, Faculty of Engineering, Chulalongkorn University
 - Specialized in applying AI/ML/DL into various domains & design Big Data platform (DB/DW)
 - Course: Data Science and Data Engineering, Data Warehousing, Natural Language Processing, etc.
- Project advisors for many companies, specialized in intelligent customer data platforms (data pipeline, model construction/deployment, ML operations)
- Certified SAS instructor
- Certified instructor for Deep Learning Institute
- Parts of NVAITC (NVIDIA AI Technology Center)



Objective: Digital Dexterity & Data Intelligence

- Be able to understand about recent digital technologies
- Be able to understand the process of applying these technologies
- Be able to understand the data and needs in your department
- Be able to plan and define KPI's for applying these technologies in your department

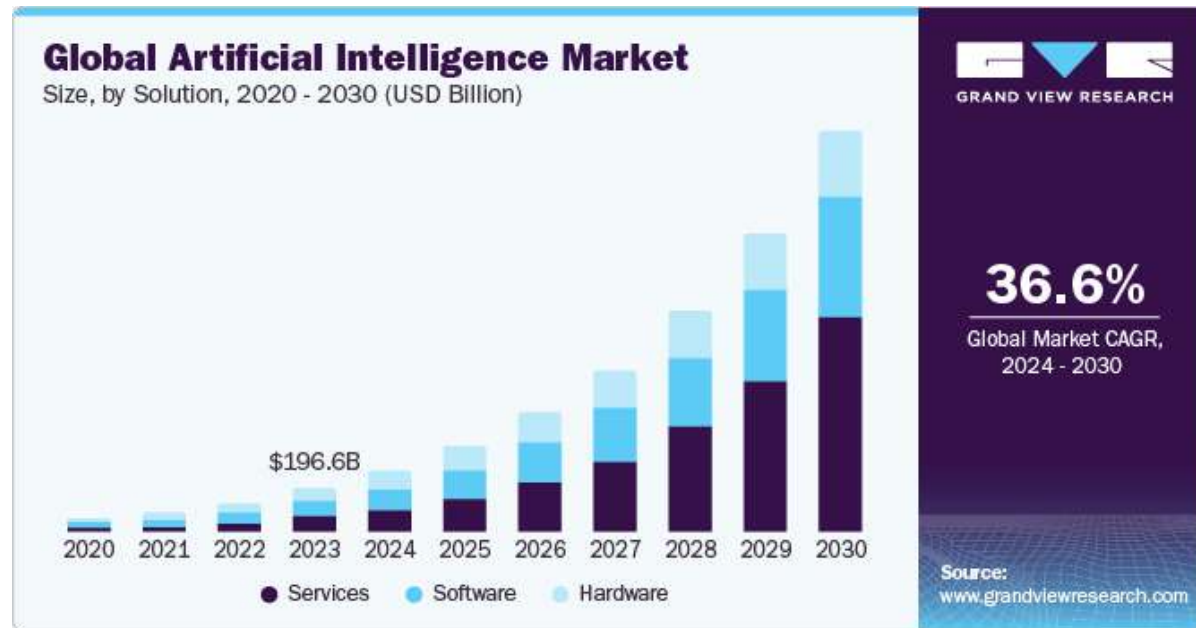
Outline

- Introduction
- Digital & AI Landscape
 1.  Document → KNOW
 2.  Table → PREDICT
 3.  Image / Video → SEE
 4.  Audio → LISTEN / SPEAK
 5.  GenAI → GENERATE
 6.  AI Coding → BUILD
 7.  Agent → ACT
- How do we deploy AI models?
 - Data
 - ML
 - Dev
 - Ops
- Group Activity
- Conclusion

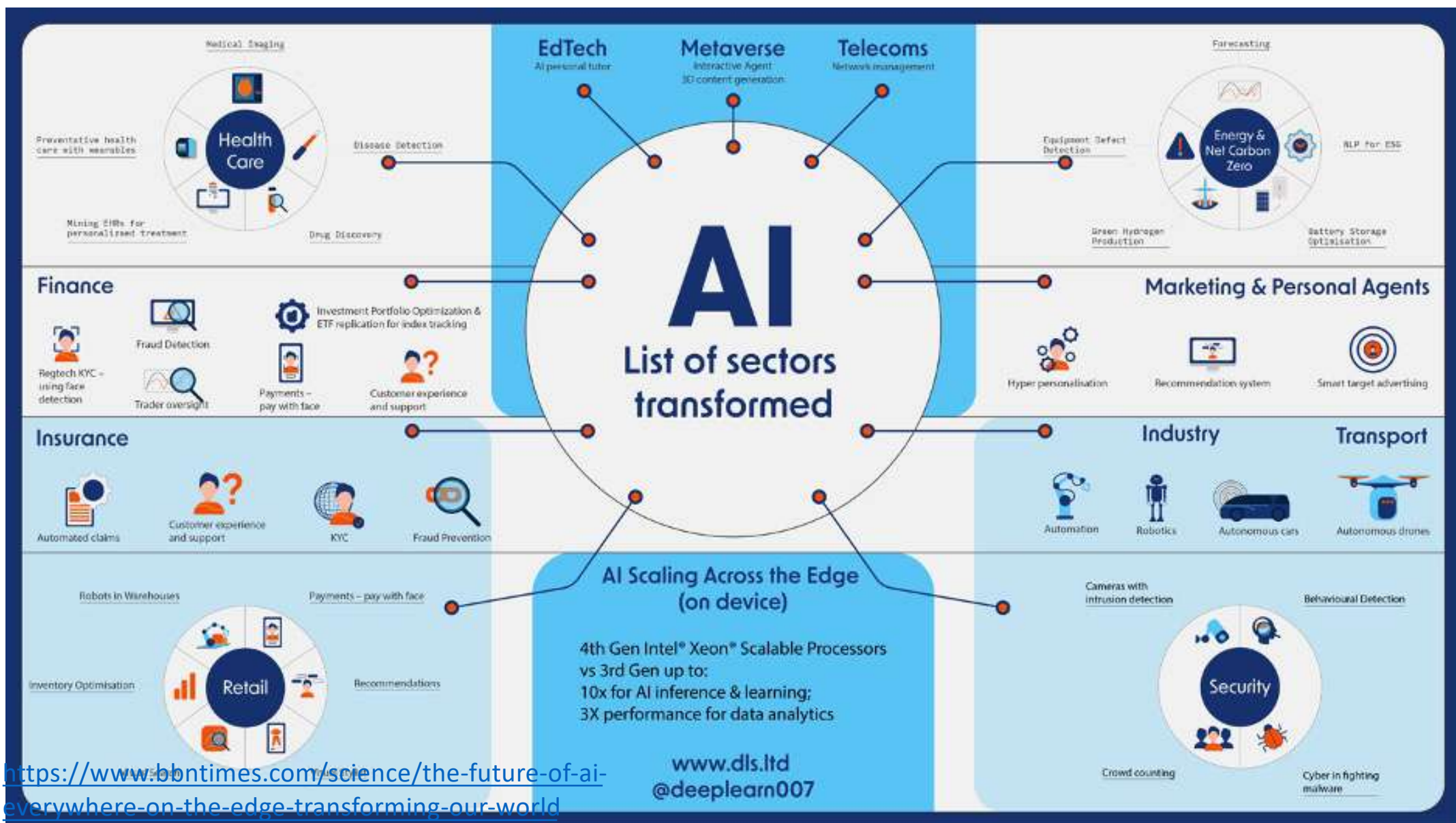
Introduction

- Nowadays it is an Era of AI, and AI is the most popular digital technology.

AI is everywhere, and technology is changing faster than ever before.



<https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market>

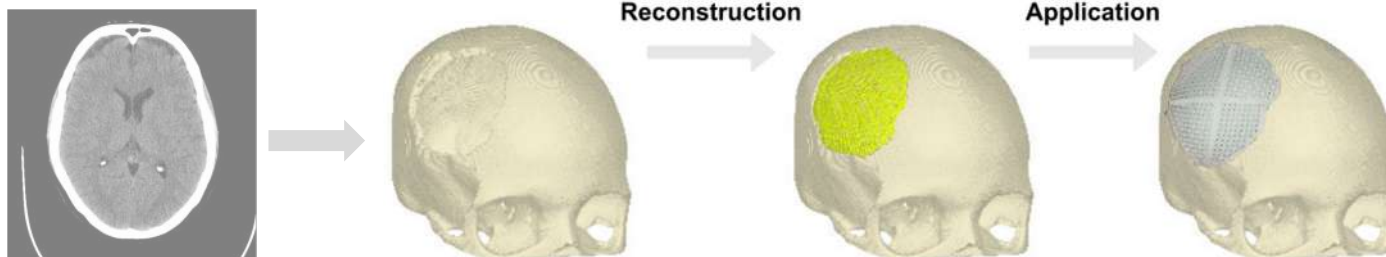
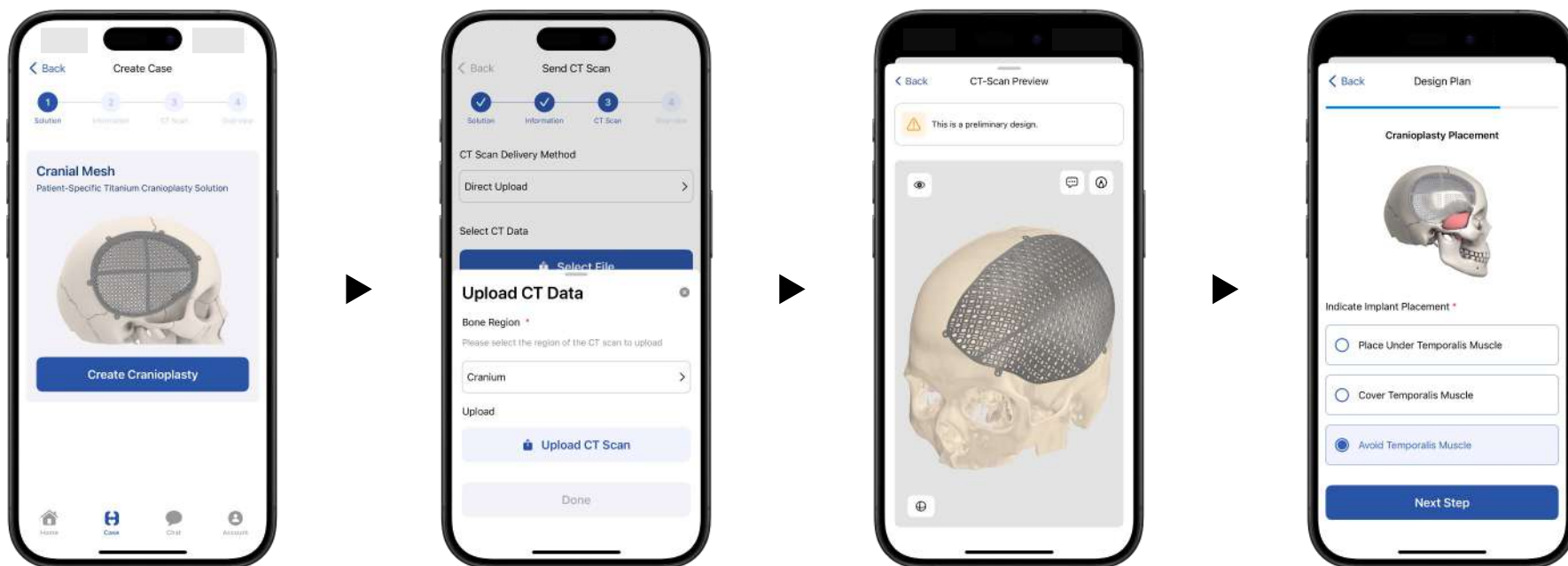


<https://www.bntimes.com/science/the-future-of-ai-everywhere-on-the-edge-transforming-our-world>



The AI feature generates the initial cranioplasty design from CT scans, making it easier for design engineers to communicate effectively with surgeons and gather precise surgical plans.

Provide design planning with the **AI-assisted design** from the uploaded CT scan



(2) 2022 - 2023

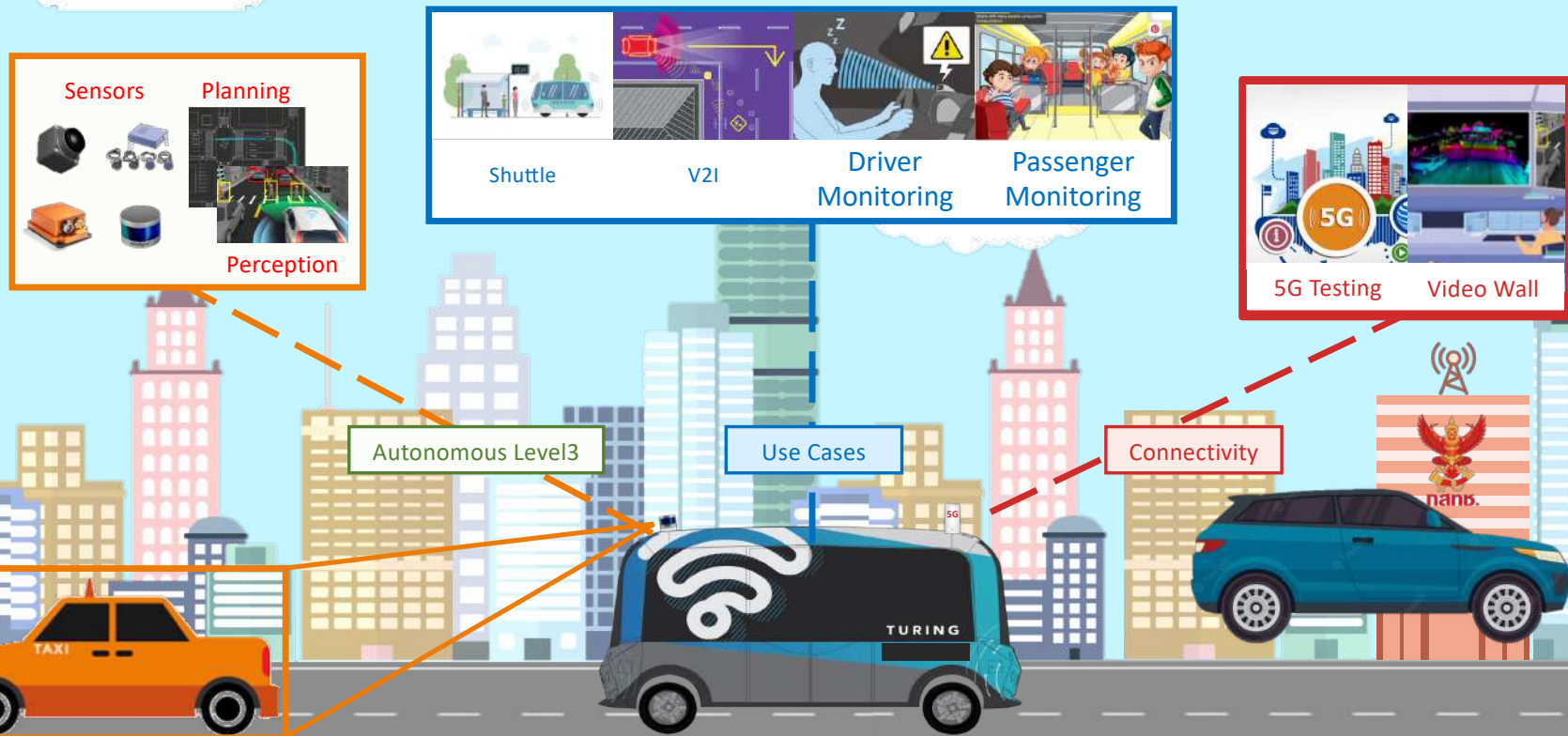


CHULA ENGINEERING
Innovation toward Sustainability | ACTING

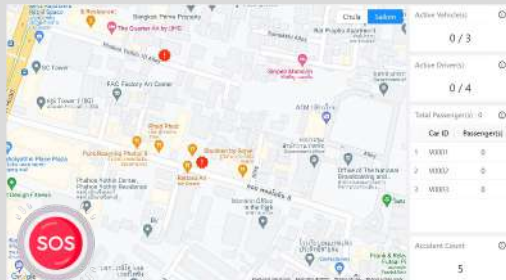


สาริตการำงาน โครงการ การทดลองการสื่อสารด้วยระบบ 5G สำหรับรถไร้คนขับ (Demo Day)

วันพุธที่ 8 กุมภาพันธ์ 2566 เวลา 13.00-16.30น.



Accident Notification



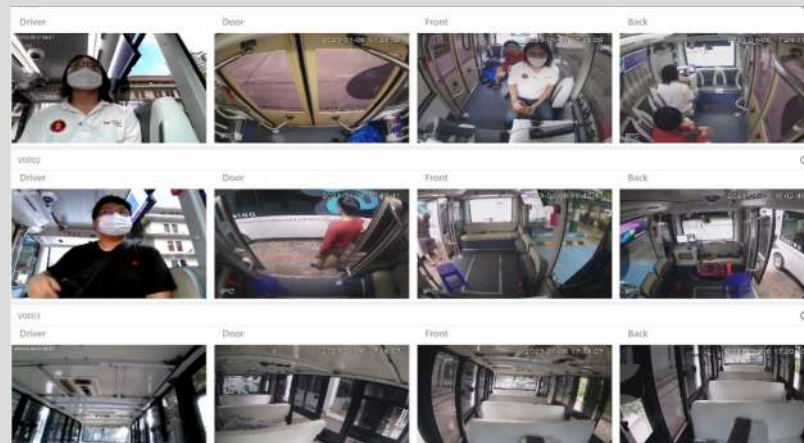
Drowsiness Detection



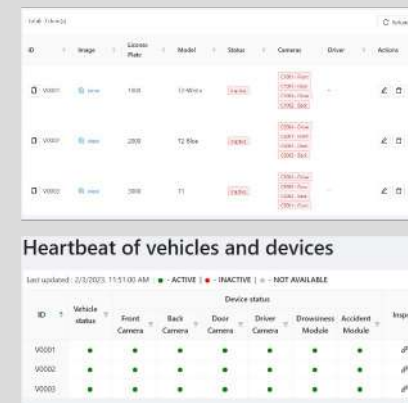
Passenger Counting



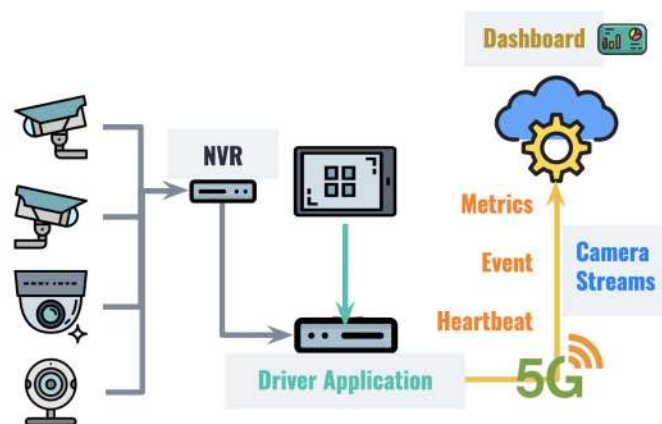
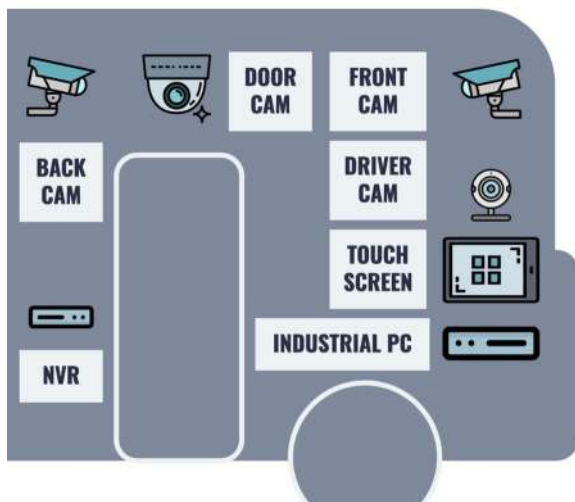
Video Streaming from 3 Autonomous Vehicles



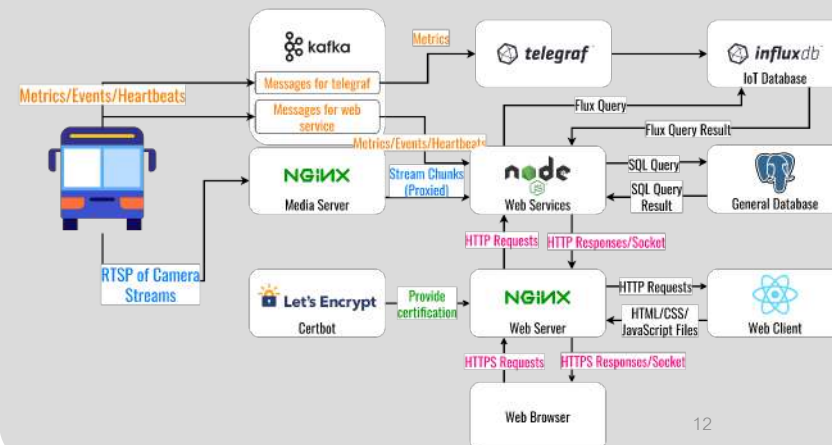
Management Web Application & Monitoring Dashboard



USE CASES



Public Server/Cloud Architecture



USE CASES: AI Modules

Passenger Counting



1. People Detection



2. Bounding Box



3. Passing Detection



4. Passenger Graph

Future Work



YOLOX
Exceeding YOLO series in 2021

An anchor-free version of YOLO: Real-time object detection

Drowsiness Detection



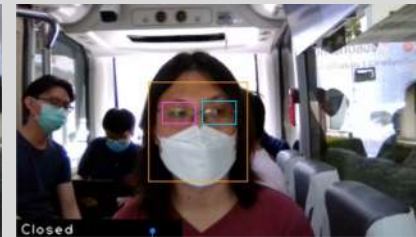
1. Brightness and Gamma Correction



2. Facial Key Point Detection with Dlib



3. Non-interested Face Filtering



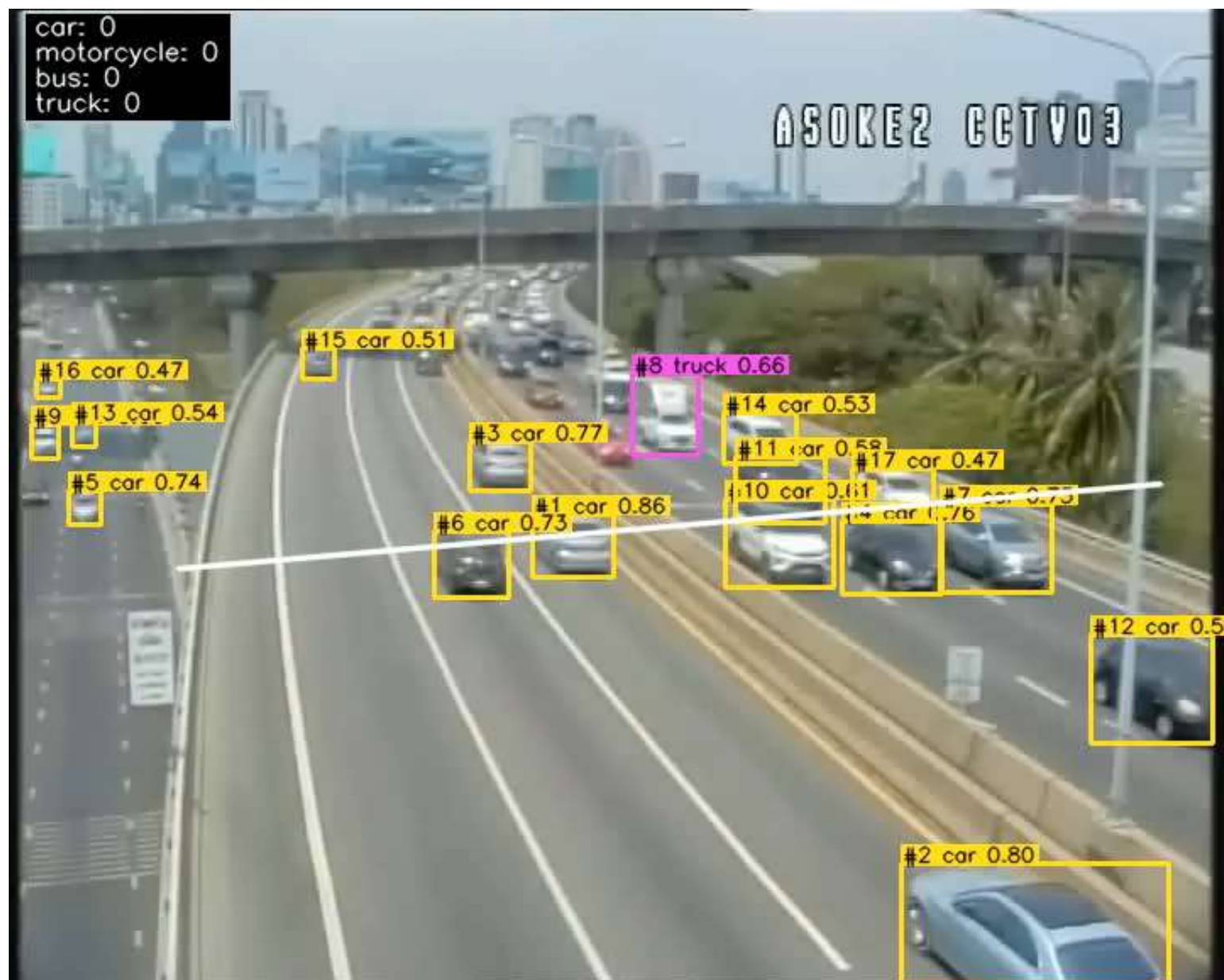
4. Detect Eyes Aspect Ratio with OpenCV

Future Work



MediaPipe

Machine learning pipelines for processing video

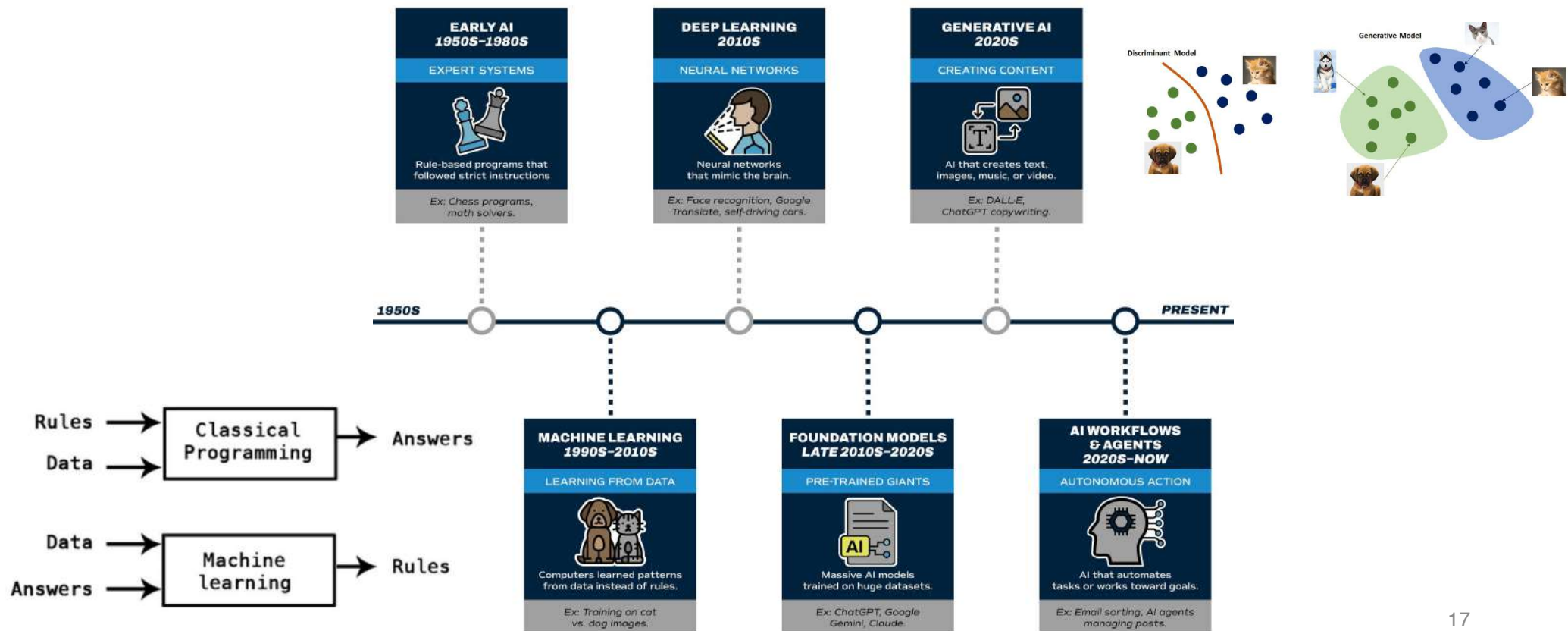


Digital Dexterity & Data Intelligence:

Digital & AI Landscape

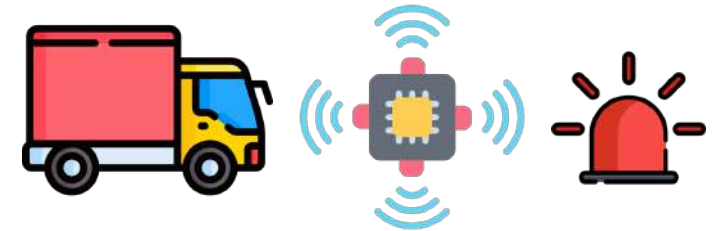
AI Landscape

🖥️ 1) Software → 📊 2) Analytics → 🧠 3) ML/DL → ✨ 4) GenAI → 🤖 5) AI Agents

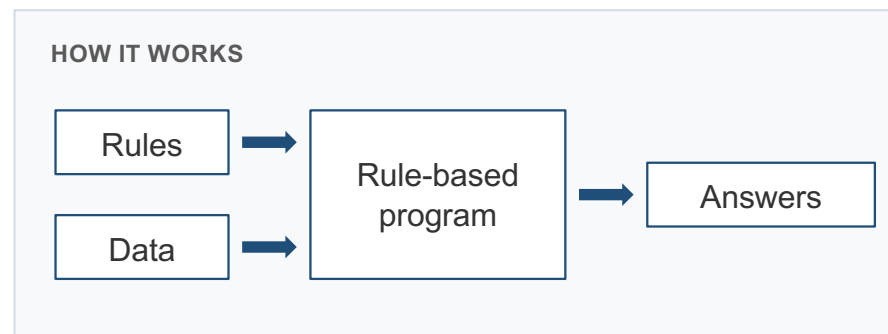




1) Traditional Software



Paperwork written into code: a person defines the rules, the computer follows them exactly.

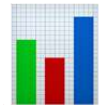


Strengths

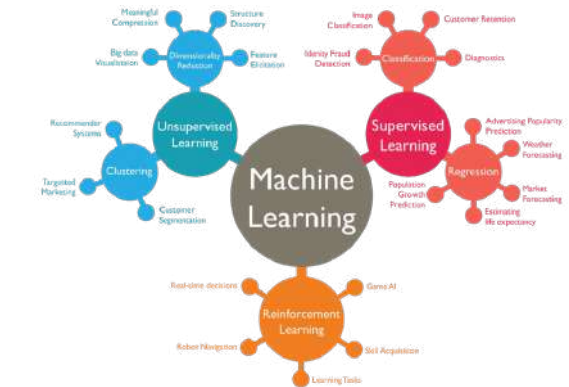
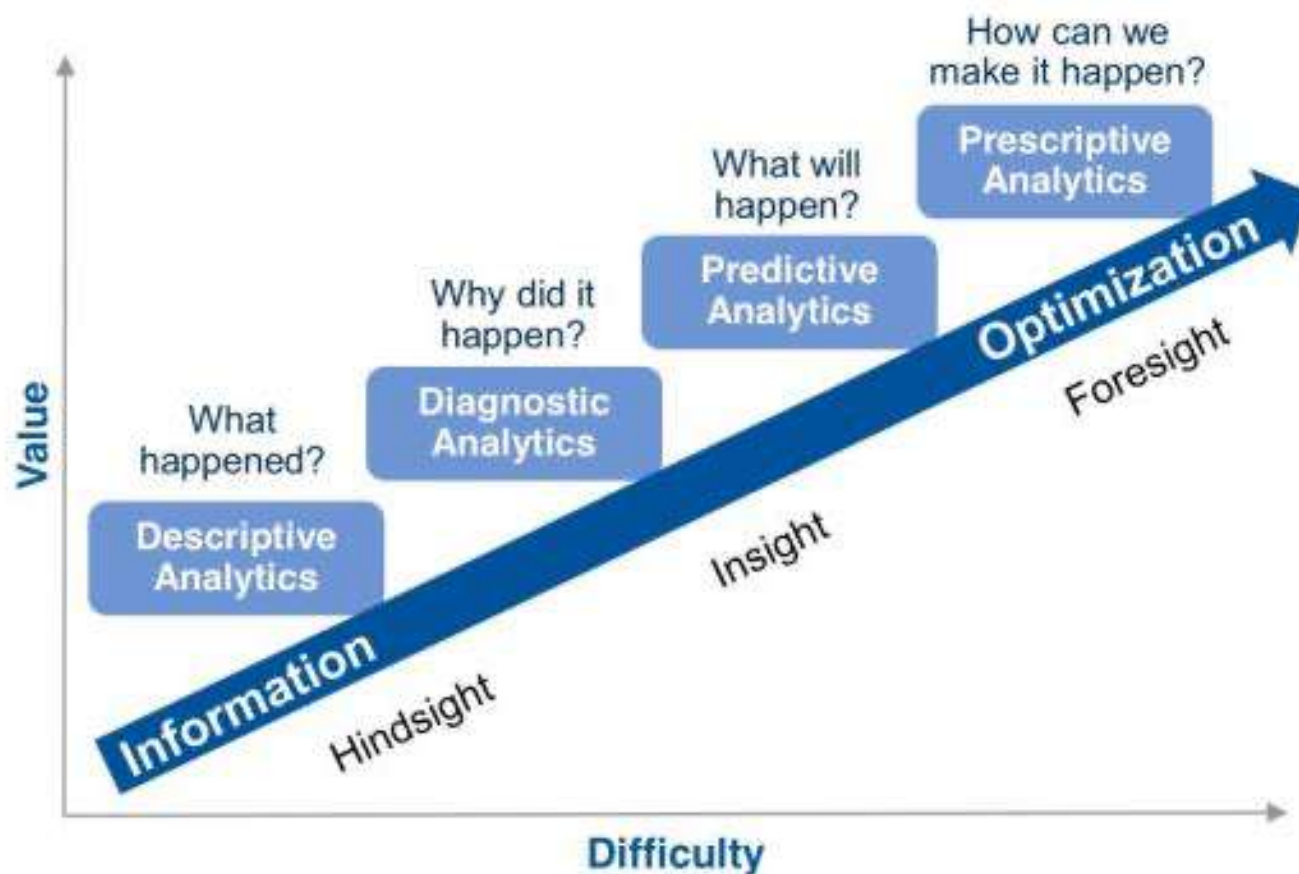
Speed, consistency and a clear audit trail at every step.

Limits

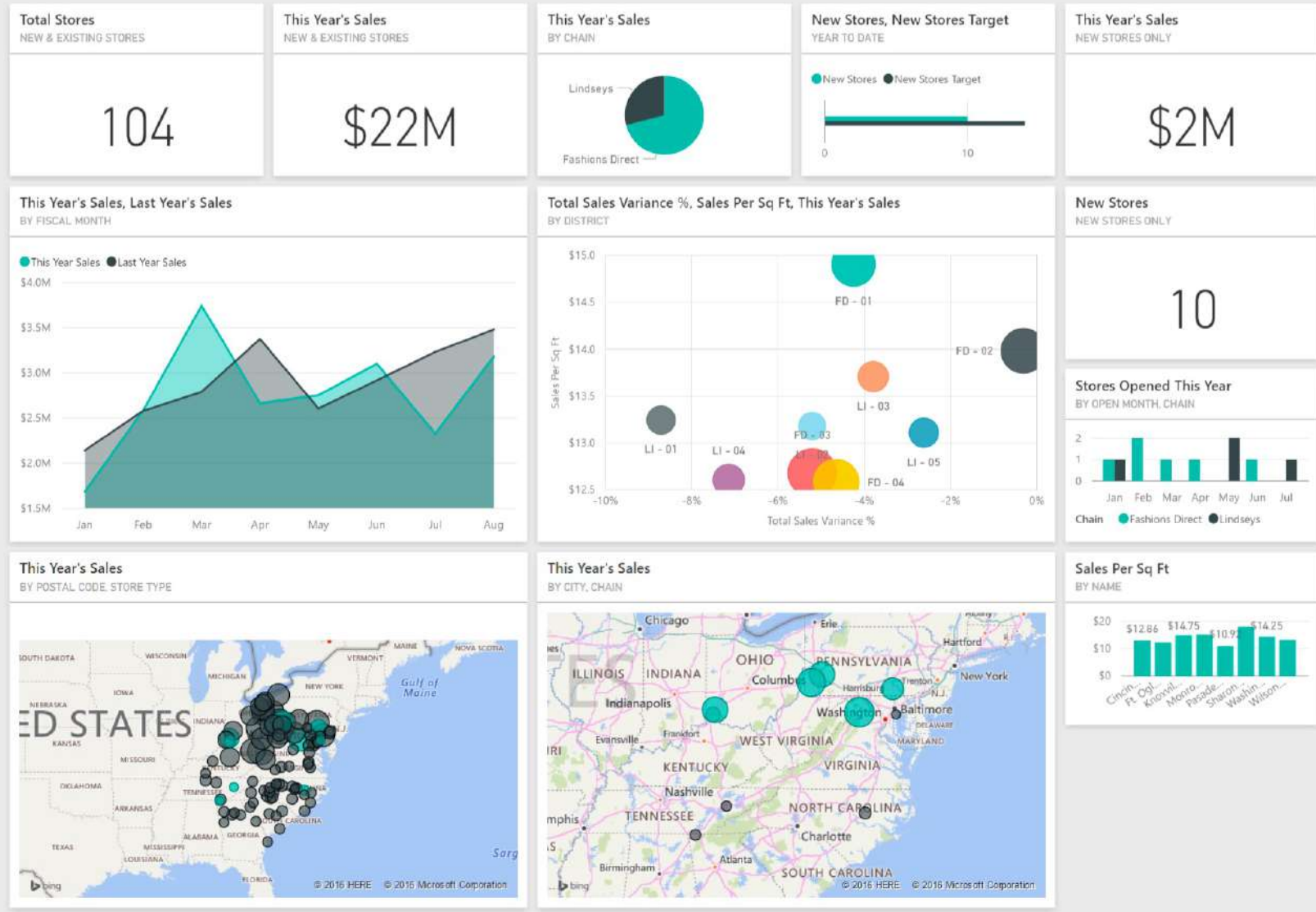
Every case must be written down first. Anything unwritten goes back to a person.



2) Data Analytics → Finding Insights



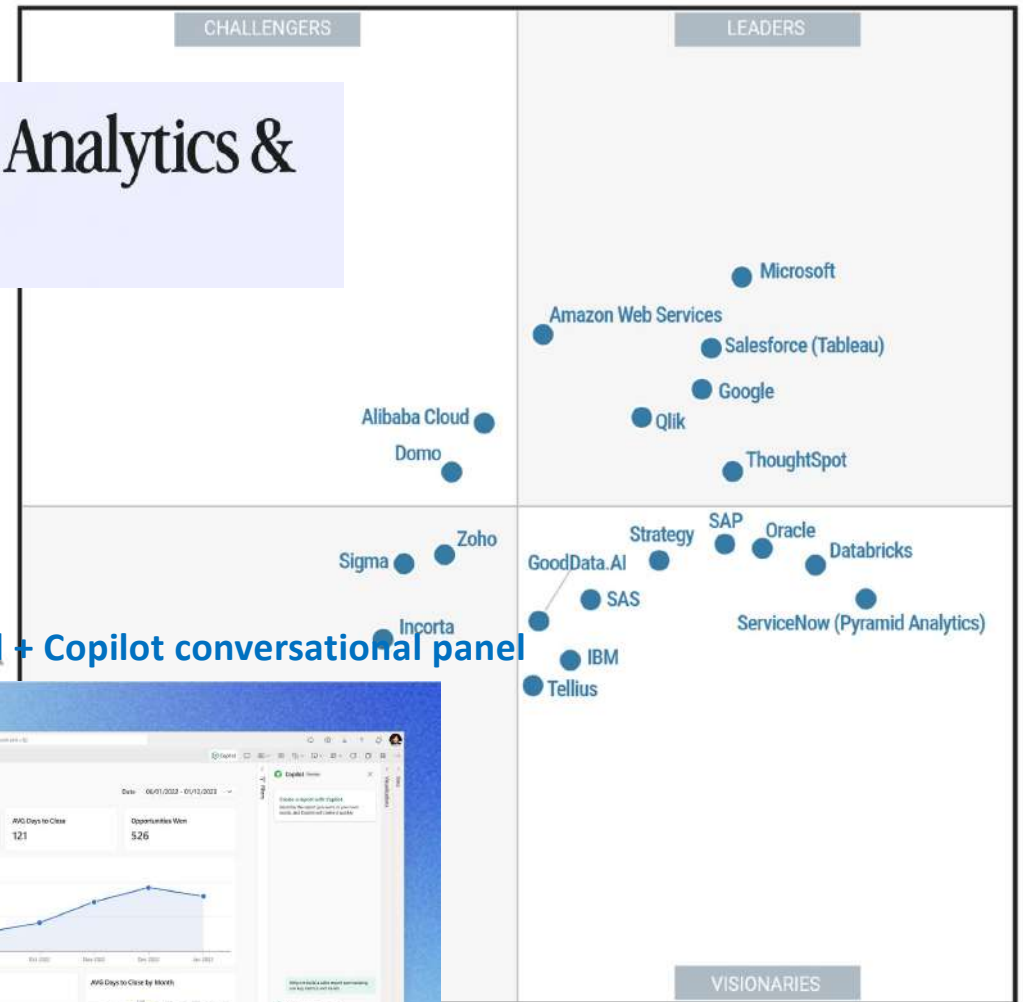
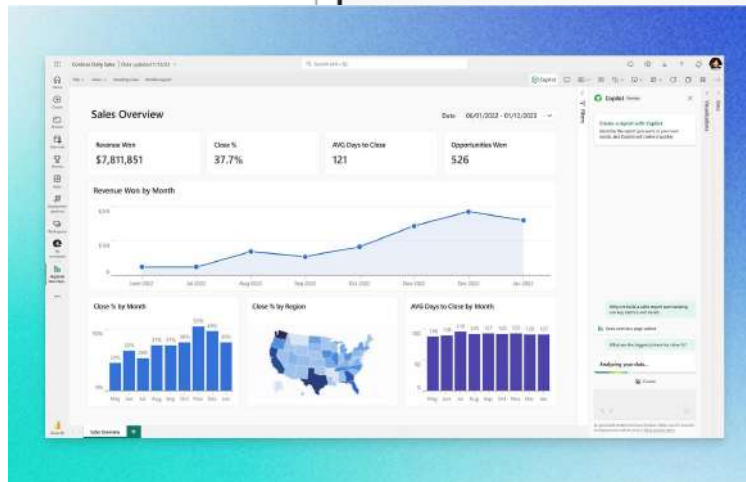
Ask a question about your data



The 2026 Gartner® Magic Quadrant™ for Analytics & Business Intelligence



Power BI dashboard + Copilot conversational panel



As of May 2026

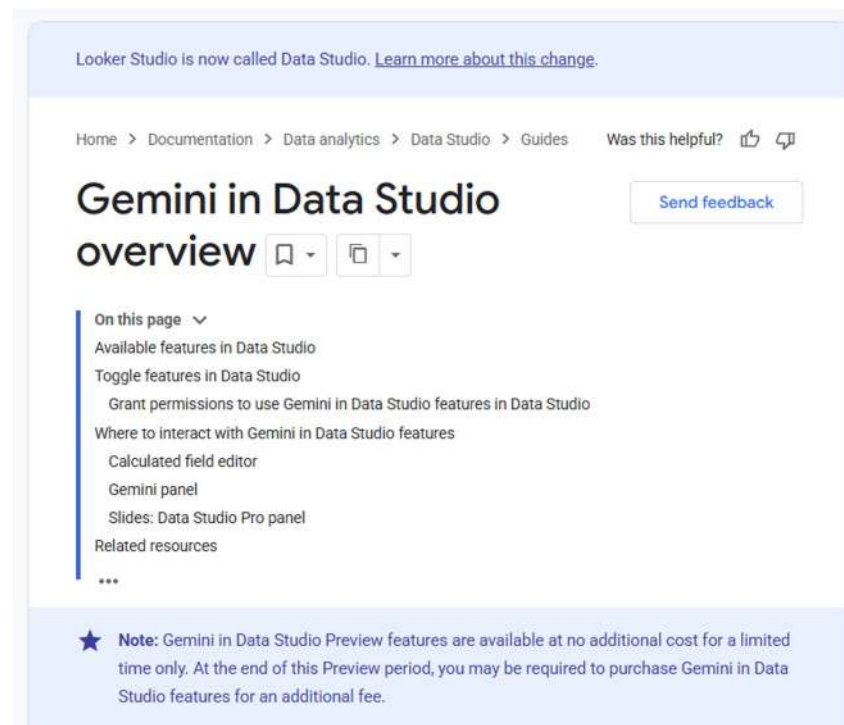
© Gartner, Inc

[Ref Link](#)

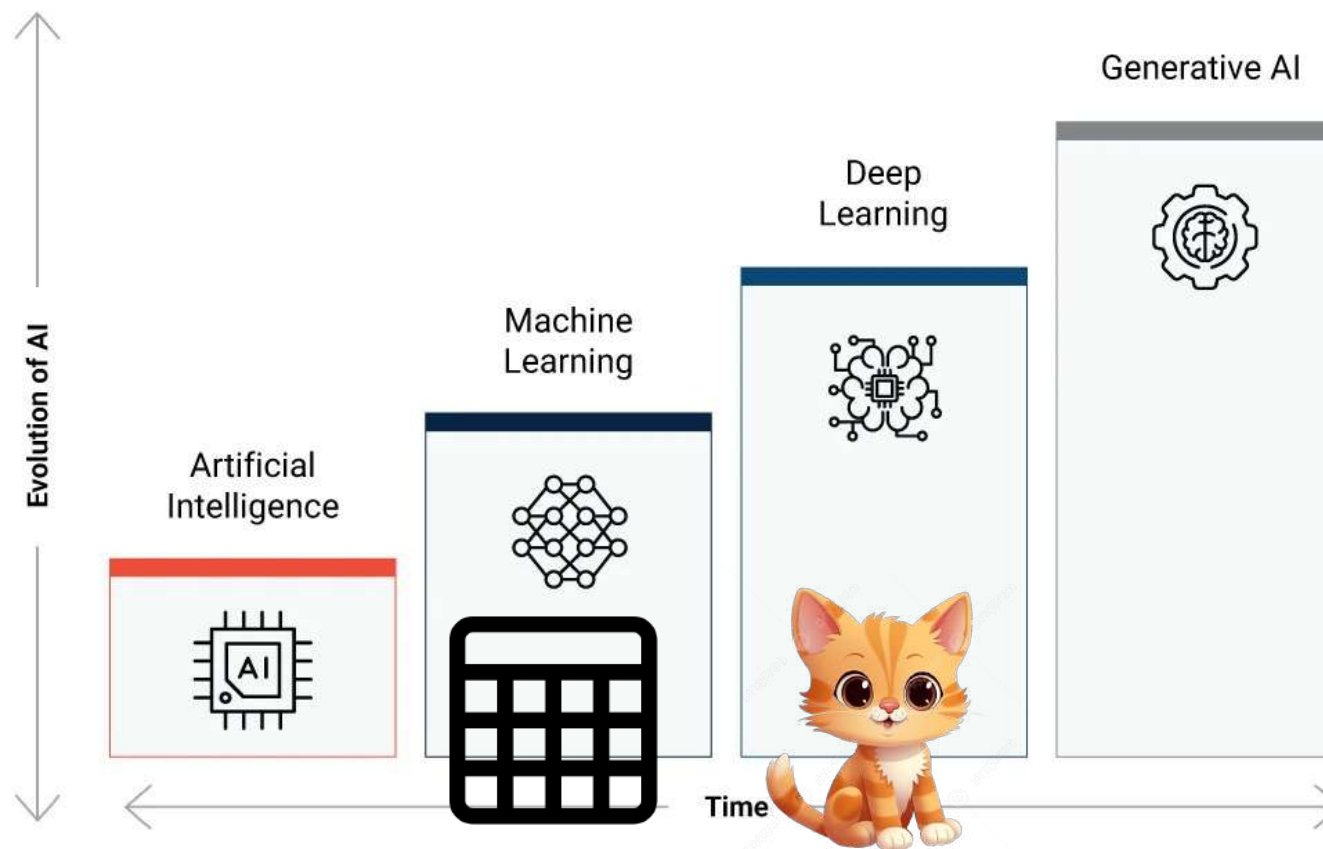
Gartner

Copilot in Google Data Studio

Gemini features inside Data Studio are currently available for a limited preview period at no additional cost, but they require **a paid Data Studio Pro subscription**.



3) ML/DL → Predict



3.1) Machine Learning (ML) → Prediction Task

Handcrafted features

Training Data



inputs				target
Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No

Testing Data

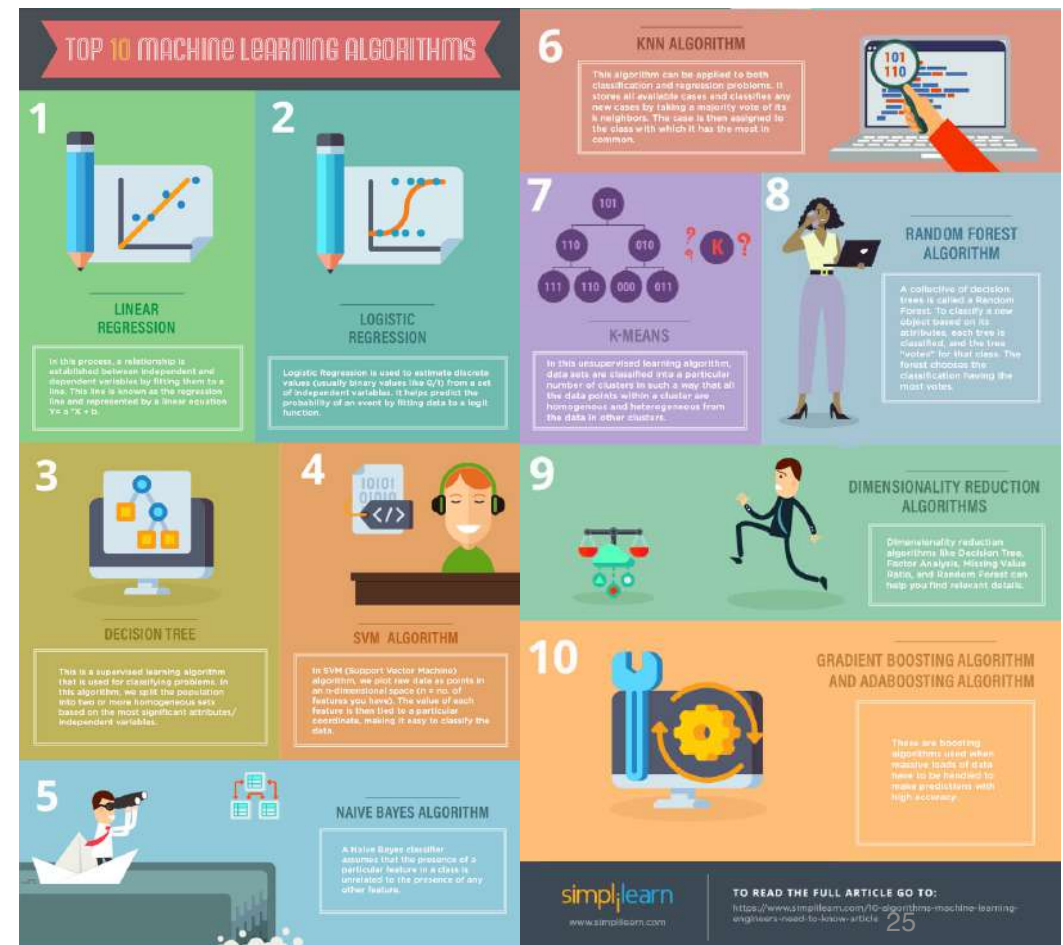


Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	?

Application: Direct Target Marketing

Prediction Algorithms

- Decision Tree
- (Logistic) Regression
- Neural Networks (NN)
- k-Nearest Neighbors (kNN)
- Support Vector Machine
- Deep Learning





Enes Gokce

Jan 10, 2020 · 20 min read · Listen



Predicting Used Car Prices with Machine Learning Techniques

Comparing Performance of Five Different ML Models

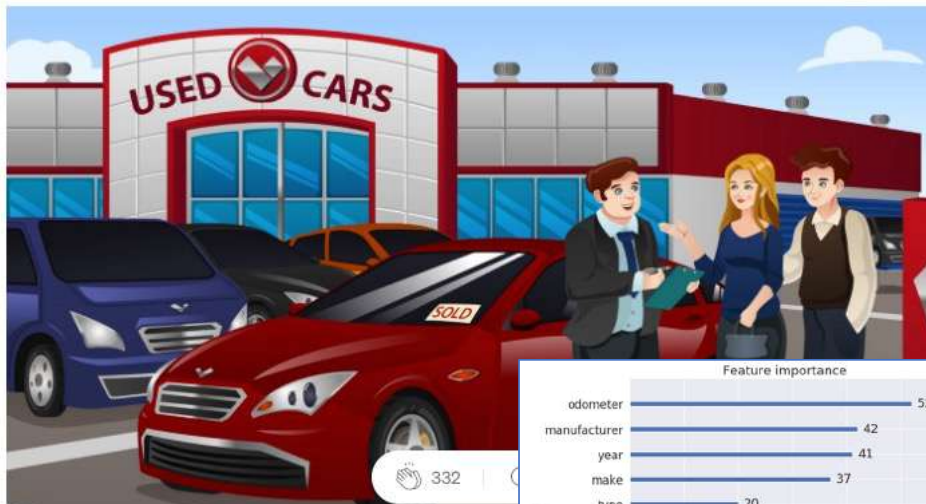


Table 15

Comparison of Model Outcomes

Measure (%) / Model	Random Forest	Ridge	XGBoc
Mean Absolute Error	2047.74	5405.78	5257.01
Mean Squared Error	15682494.73	57436987.51	57438444.92
RMSE	3960.11	7578.72	7578.82
			9698.92
			9698.92

Figure 21: XGBoost Plot for Feature Importance

Search

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Register

PSY · UPDATED 3 YEARS AGO

27

New Notebook

Download (590 kB)

Bank Direct Marketing

The data is related with direct bank marketing campaigns



Data Card Code (4) Discussion (0)

About Dataset

Citation Request:

This dataset is public available for research. The details are described in (Moro et al., 2011).

Please include this citation if you plan to use this database:

[Moro et al., 2011] S. Moro, R. Laureano and P. Cortez, Using Data Mining for Bank Direct Marketing: An Application of the CRISP-DM Methodology.

In P. Novais et al. (Eds.), Proceedings of the European Simulation and Modelling Conference - ESM'2011, pp. 117-121, Guimarães, Portugal, October, 2011. EUROSIS.

Available at: [pdf] <http://hdl.handle.net/1822/14838>

Usability

10.00

License

Data Files Original Authors

Expected update frequency

Never

Banking

Marketing

bank-full.csv (4.61 MB)

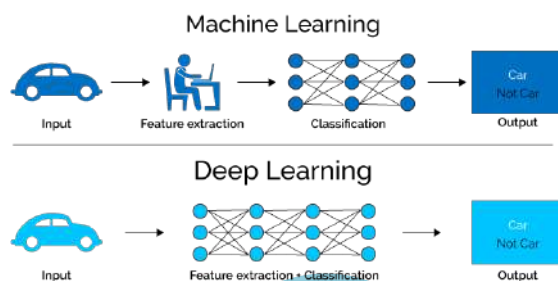
Detail Compact Column

17 of 17 columns

contact	# day	month	duration	# campaign	# pdays	# previous	outcome	y
known	5	may	261	1	-1	0	unknown	no
known	5	may	151	1	-1	0	unknown	no
known	5	may	76	1	-1	0	unknown	no
known	5	may	92	1	-1	0	unknown	no
known	5	may	150	1	-1	0	unknown	no
known	5	may	130	1	-1	0	unknown	no
known	5	may	217	1	-1	0	unknown	no
known	5	may	380	1	-1	0	unknown	no
known	5	may	58	1	-1	0	unknown	no
known	5	may	55	1	-1	0	unknown	no
known	5	may	259	1	-1	0	unknown	no

<https://www.kaggle.com/datasets/psymor/bank-direct-marketing/discussion>

3.2) Deep Learning (DL)



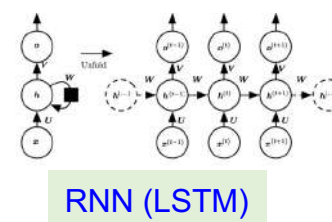
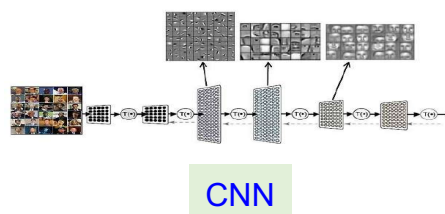
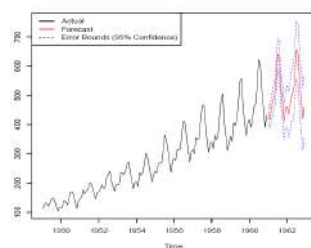
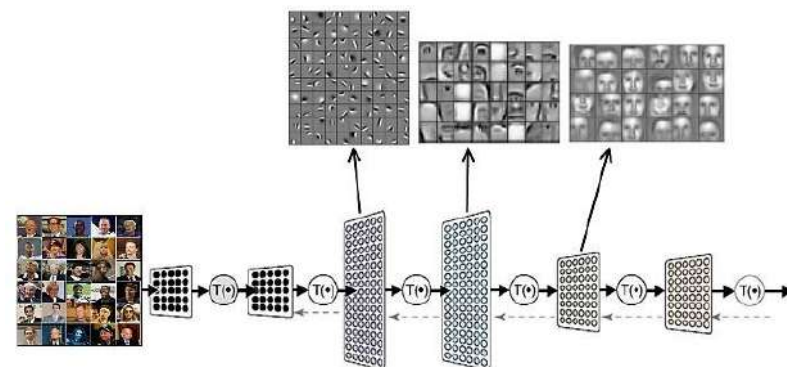
Speech
Recognition



Computer
Vision



Natural Language
Processing





English | 简体中文



Ultralytics YOLOv8 is a cutting-edge, state-of-the-art (SOTA) model that builds upon the success of previous YOLO versions and introduces new features and improvements to further boost performance and flexibility. YOLOv8 is designed to be fast, accurate, and easy to use, making it an excellent choice for a wide range of object detection and tracking, instance segmentation, image classification and pose estimation tasks.

Models

YOLOv8 [Detect](#), [Segment](#) and [Pose](#) models pretrained on the [COCO](#) dataset are available here, as well as YOLOv8 [Classify](#) models pretrained on the [ImageNet](#) dataset. [Track](#) mode is available for all Detect, Segment and Pose models.

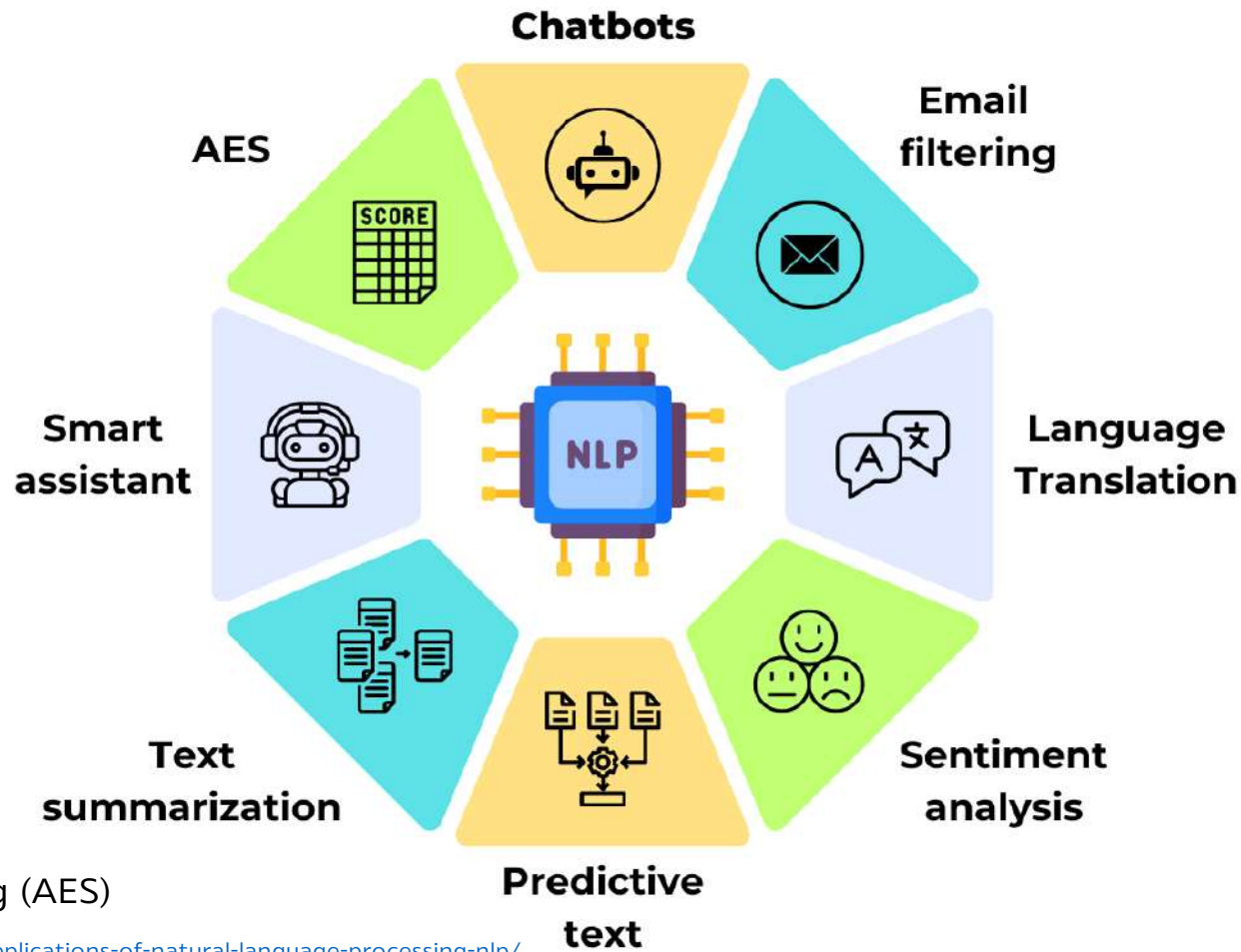


All [Models](#) download automatically from the latest Ultralytics [release](#) on first use.

<https://github.com/ultralytics/ultralytics>

Top 8 Applications of Natural Language Processing (NLP)

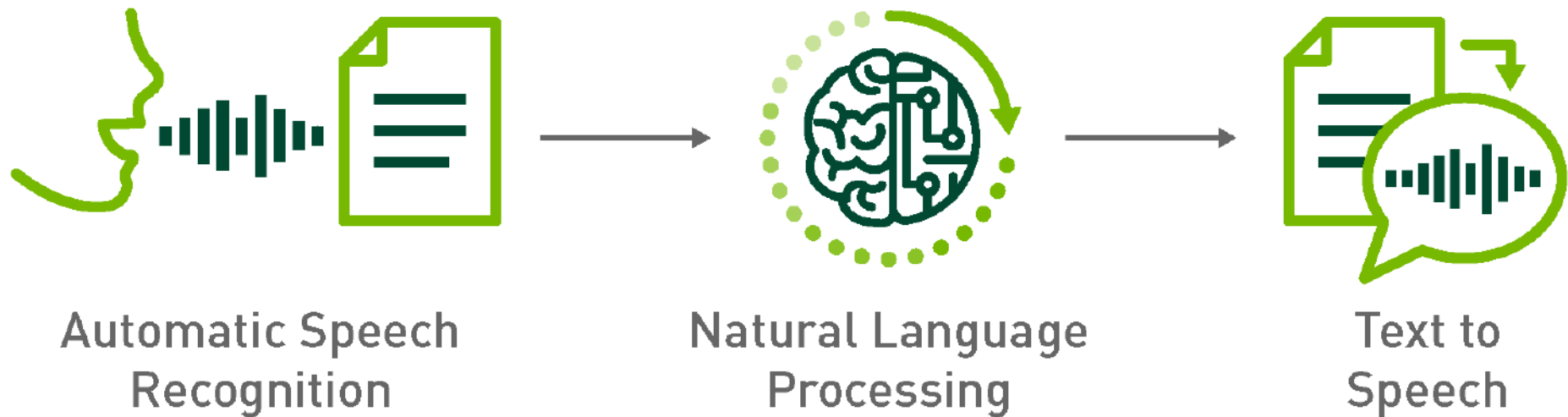
Applications of Natural Language Processing



Automated Essay Scoring (AES)

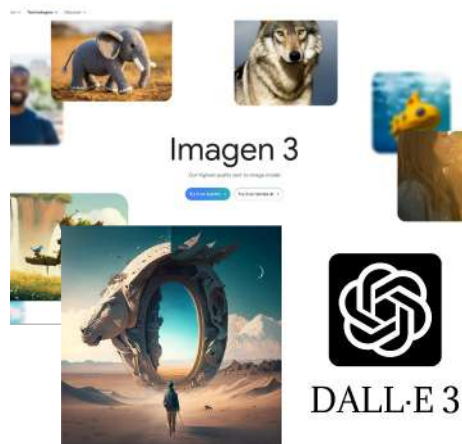
<https://eastgate-software.com/top-8-applications-of-natural-language-processing-nlp/>

Automatic Speech Recognition (ASR) & Text-to-Speech (TTS)

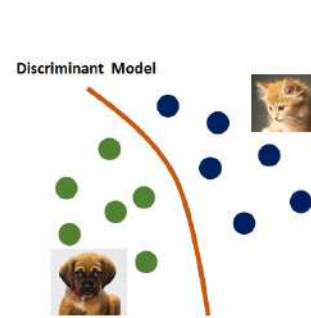
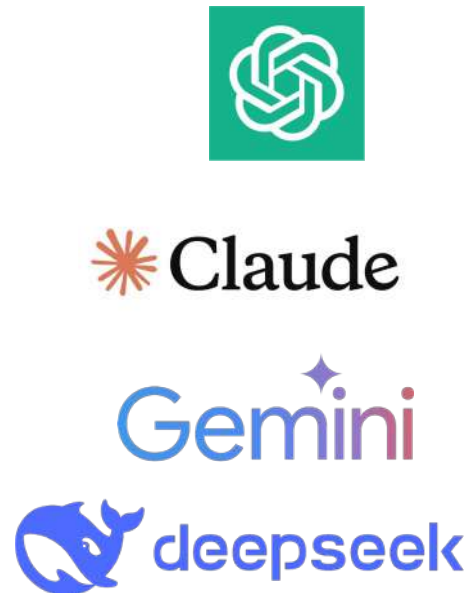


4) GenAI

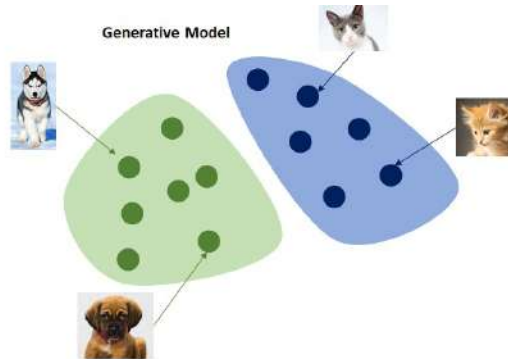
1) Image



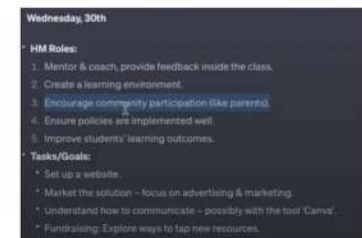
2) Text



3) Video



4) Multimodal



Prompt Engineering Techniques



Zero-shot
prompting



Few-shot prompting
or in-context
learning



Chain-of-thought
prompting



Tree-of-thought
prompting



Iterative
refinement



Feedback
loops



Prompt
chaining



Role-playing



Maieutic
prompting

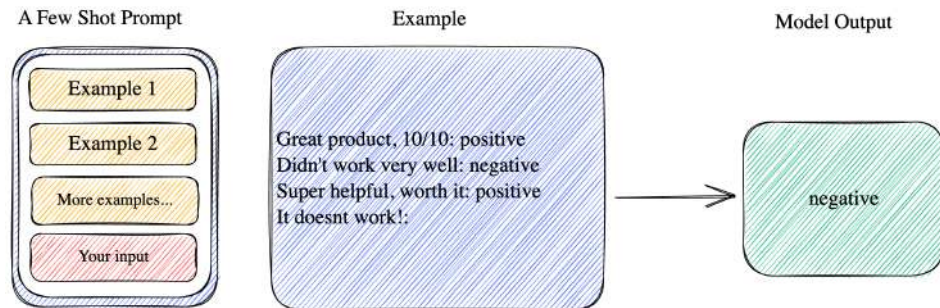


Complexity-based
prompting

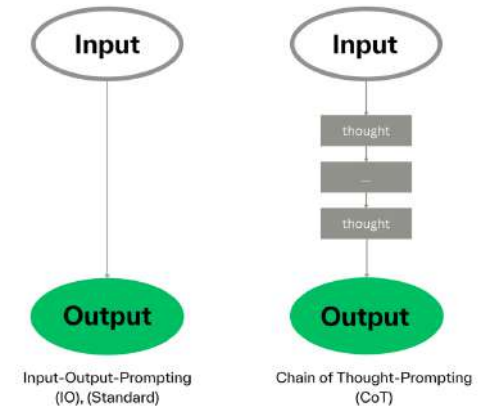
servicenow

<https://www.servicenow.com/ai/what-is-prompt-engineering.html>

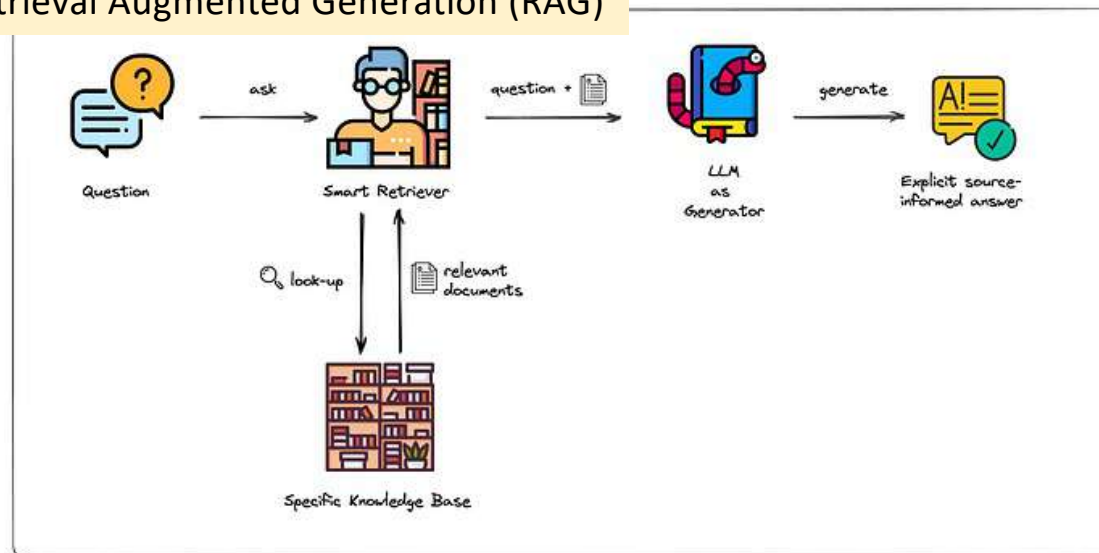
1) Few-Shot Prompting



2) Chain-of-Thoughts (CoT)



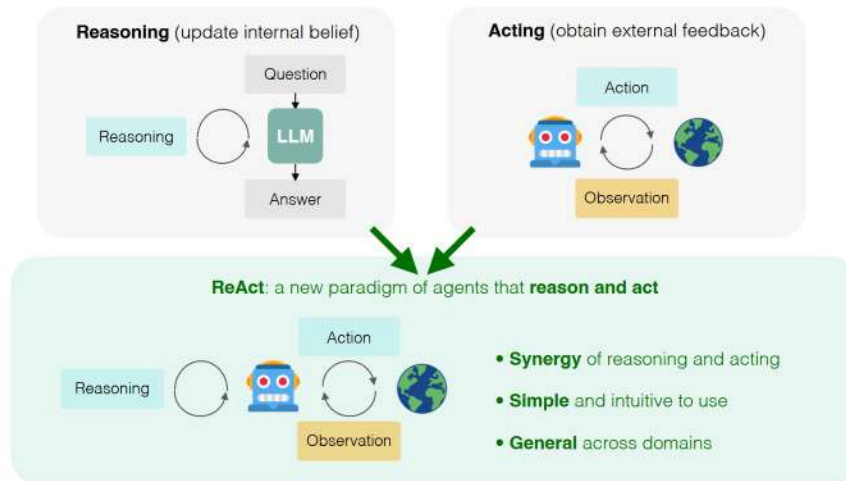
3) Retrieval Augmented Generation (RAG)





5) AI Agents

- AI is no longer just answering questions—it can understand instructions and take actions – **Agentic AI (think + code)**.
- For example, ChatGPT → **Codex = Coding + Execution**



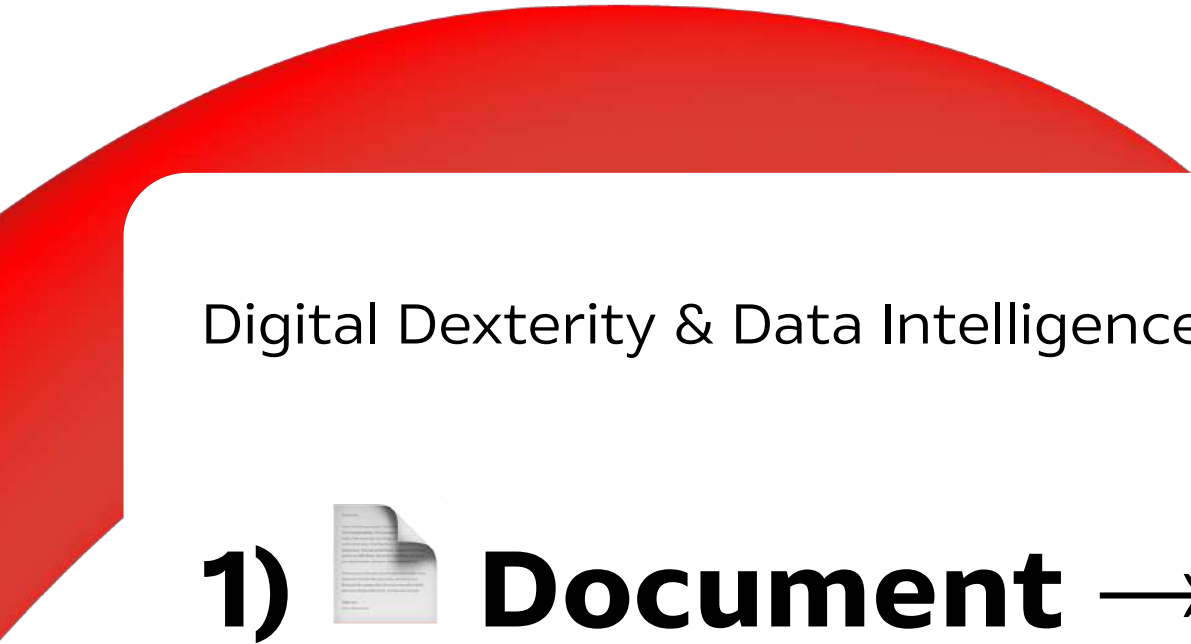
20



What can AI do?

 1) Software →  2) Analytics →  3) ML/DL →  4) GenAI →  5) AI Agents

1.  Document → KNOW
2.  Table → PREDICT
3.  Image / Video → SEE
4.  Audio → LISTEN / SPEAK
5.  GenAI → GENERATE
6.  AI Coding → BUILD
7.  Agent → ACT

A large, thick red shape on the left side of the slide, resembling a stylized arch or a thick bracket that curves from the top left towards the center.

Digital Dexterity & Data Intelligence:

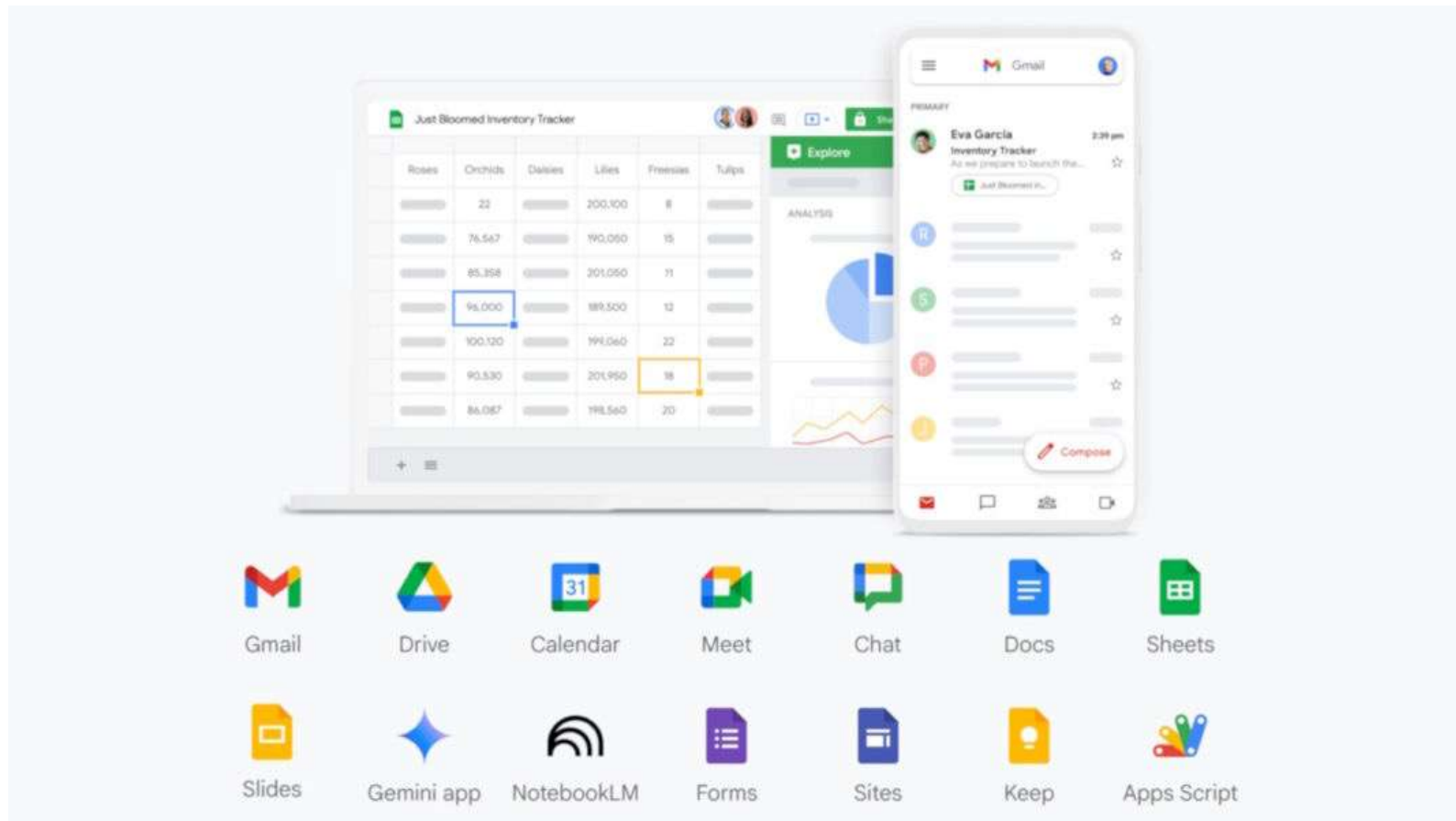
1)  **Document** → **KNOW**

A small red parallelogram shape located on the right side of the slide.

Lab 1.1 (food_prices.csv)

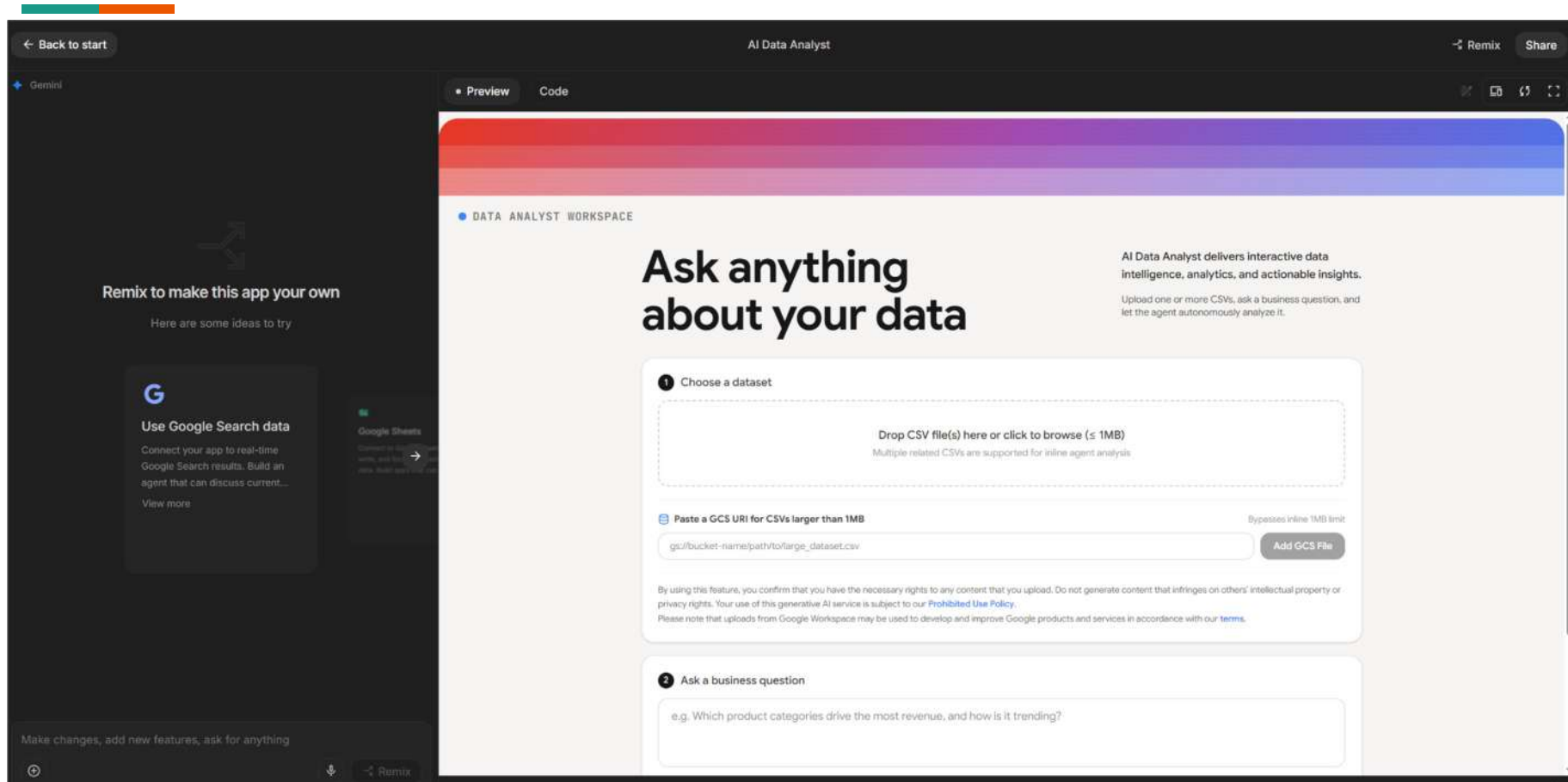


Gemini for Google Workspace

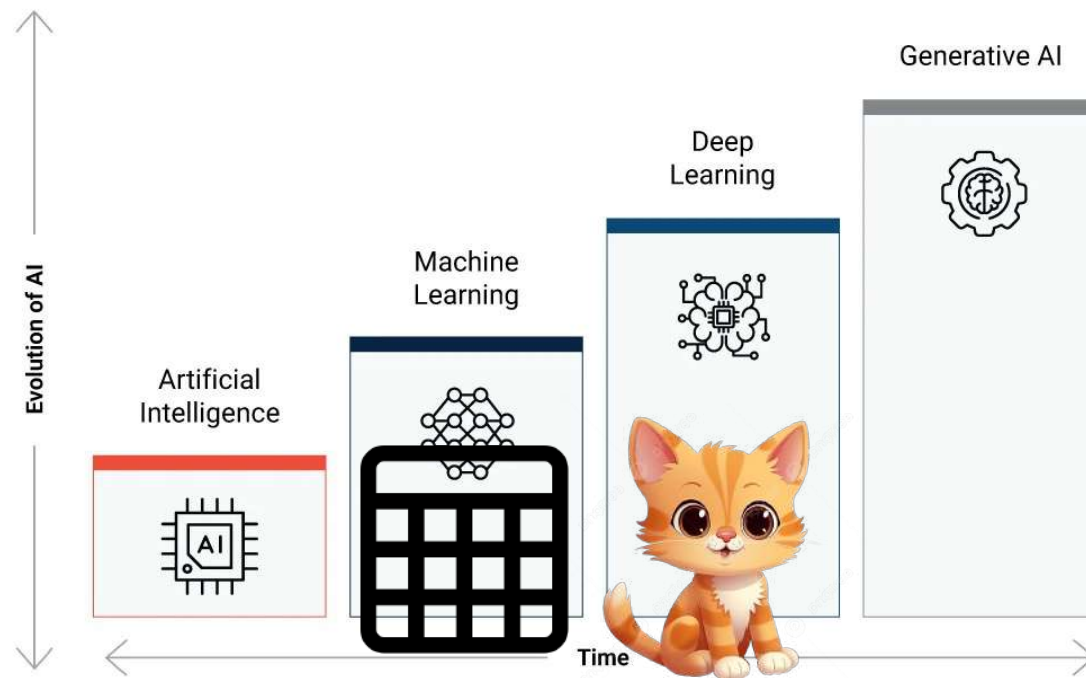


Lab 1.2 (food_prices.csv)

Google AI Studio - AI Data Analyst



2) Table → PREDICT



Machine Learning (ML) → Prediction Task

Handcrafted features

Training Data



inputs				target
Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No

Testing Data

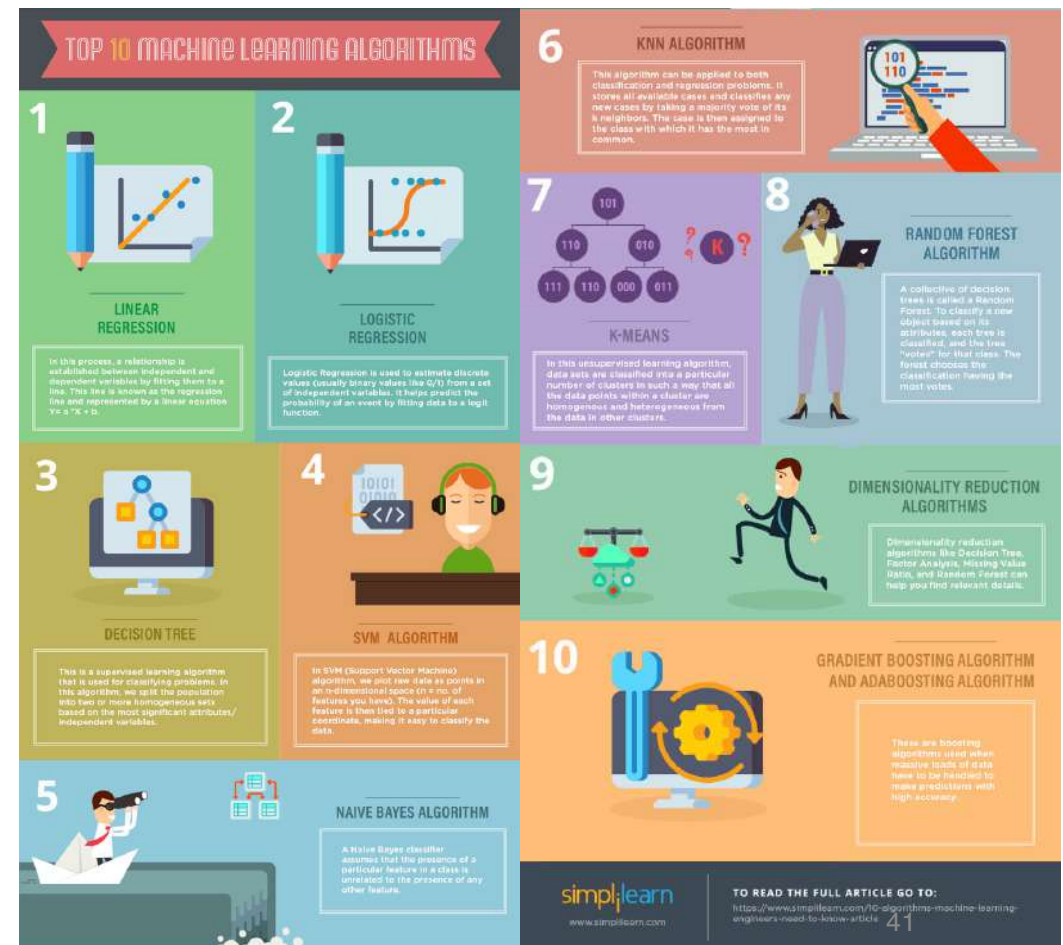


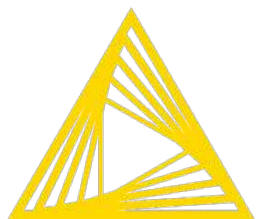
Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	?

Application: Direct Target Marketing

Prediction Algorithms

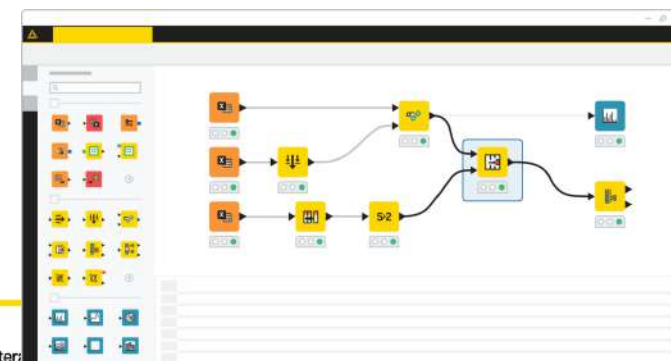
- Decision Tree
- (Logistic) Regression
- Neural Networks (NN)
- k-Nearest Neighbors (kNN)
- Support Vector Machine
- Deep Learning





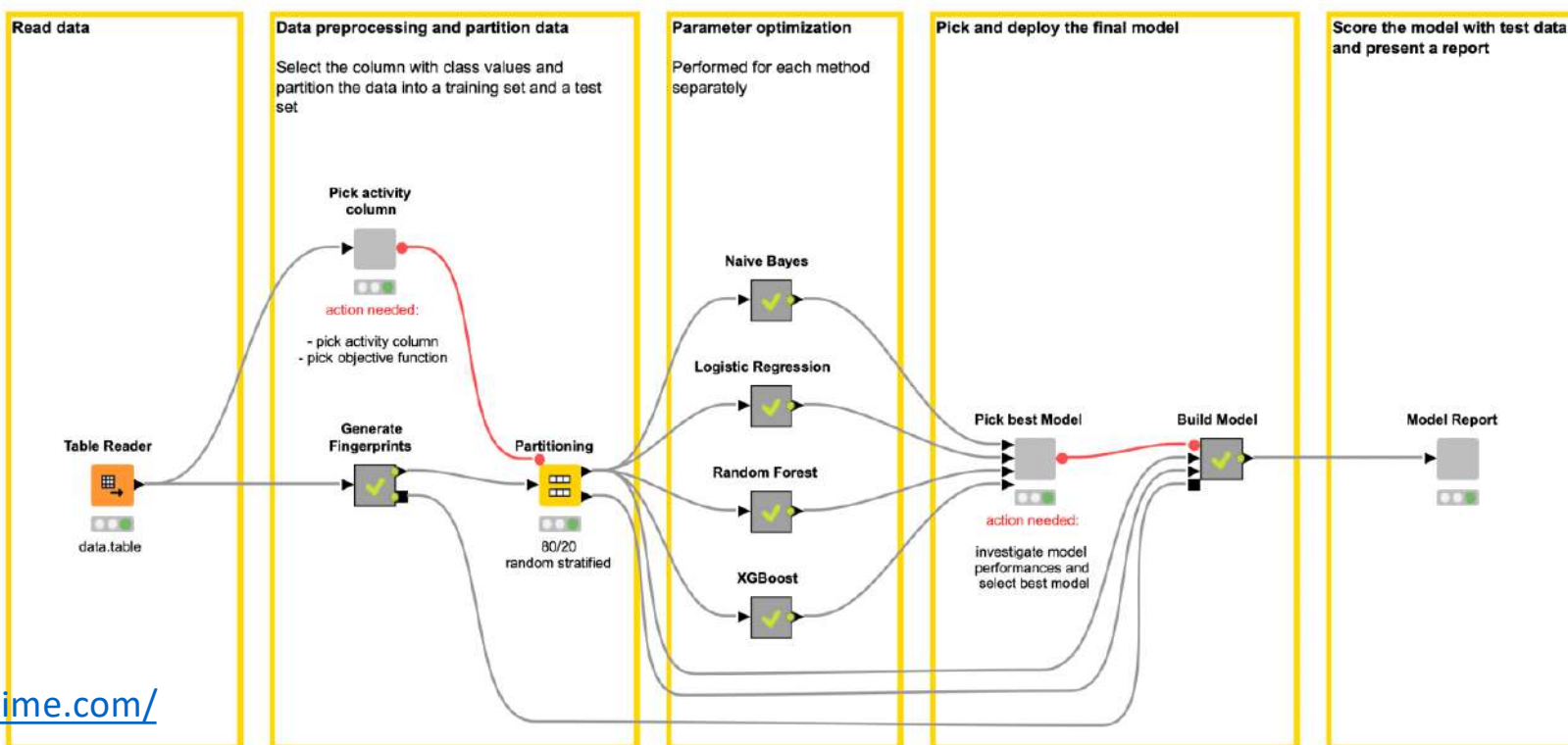
Open for Innovation

KNIME



Model Selection with Integrated Deployment

This workflow deploys an advanced parameter optimization protocol with four machine learning methods. The model performances can be compared in the interactive component. Based on the selection in the interactive view, the selected model is build and automatically deployed. Additionally, a model report is generated.



<https://www.knime.com/>

Google Colaboratory with Gemini



Lab 2 (bank.csv)

BankCustomerPrediction.ipynb

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

⋮

📄

⏪

🔍

🔗

📁

📊

[2] ✓

▶

```
1 import pandas as pd
2
3 # Load the bank.csv file into a DataFrame, allowing pd.read_csv to infer the
  comma delimiter
4 df = pd.read_csv('/content/bank.csv')
```

[3] ✓ 0s

▶

```
1 df.head()
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign
0	30	unemployed	married	primary	no	1787	no	no	cellular	19	oct	79	1
1	33	services	married	secondary	no	4789	yes	yes	cellular	11	may	220	1
2	35	management	single	tertiary	no	1350	yes	no	cellular	16	apr	185	1
3	30	management	married	tertiary	no	1476	yes	yes	unknown	3	jun	199	4
4	59	blue-collar	married	secondary	no	0	yes	no	unknown	5	may	226	1

▼

การแก้ไข Warning และ Bug

[4] ✓ 15s

▶

```
1 # Install a Thai font for matplotlib
2 # Removed problematic font installations (matplotlib-fontconfig and
  THSarabunNew.zip) as they failed.
3 !pip install -q thai-segmenter
4 !pip install -q fonttools
5
6 import matplotlib.pyplot as plt
7 import matplotlib.font_manager as fm
8 import os
9
10 # Try to find and set a common Thai font if available, or fall back to default
11 # Note: TH Sarabun New might not be available by default in all Colab
```

Gemini

🗨 ⚙️ 👤 Share 🖼

✓ RAM 100% Disk 100%

Hello, Peerapon

How can I help you today?

Display summary statistics of the DataFrame

Check for missing values in the dataset

Visualize the distribution of the 'age' column

What can I help you build?

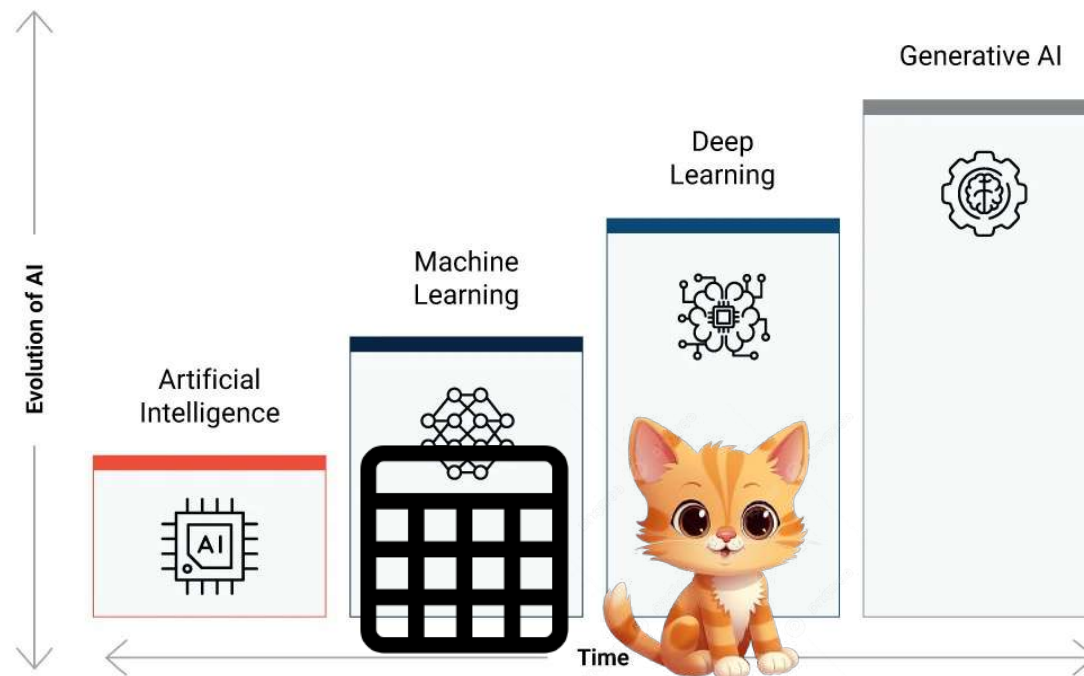
+ 📄

Gemini 2.5 Flash ▶

{ } Variables 📄 Terminal

Python 3

3) Image / Video → SEE



<https://github.com/ultralytics/ultralytics>



Lab 3.1 (food_defects, polyp_detect)



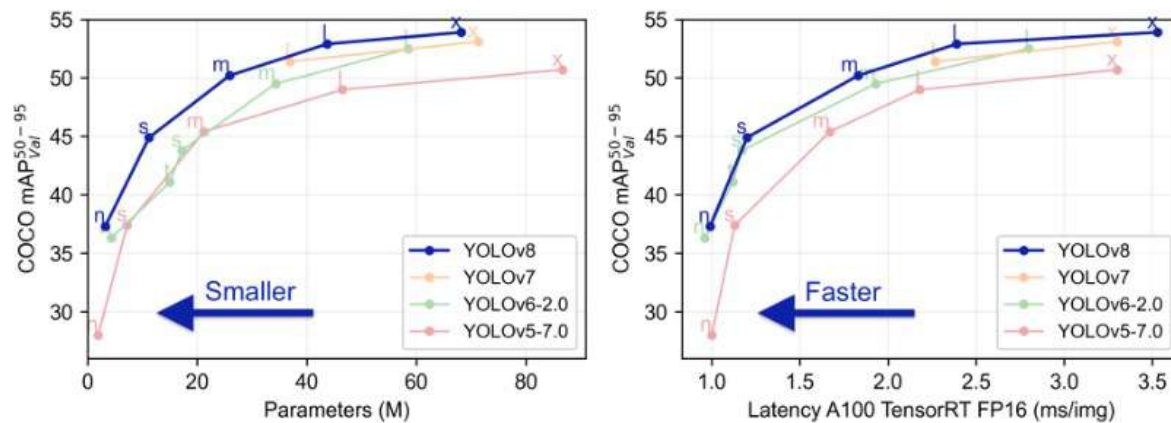
Ultralytics YOLOv8 is a cutting-edge, state-of-the-art (SOTA) model that builds upon the success of previous YOLO versions and introduces new features and improvements to further boost performance and flexibility. YOLOv8 is designed to be fast, accurate, and easy to use, making it an excellent choice for a wide range of object detection and tracking, instance segmentation, image classification and pose estimation tasks.

Models

YOLOv8 [Detect](#), [Segment](#) and [Pose](#) models pretrained on the [COCO](#) dataset are available here, as well as YOLOv8 [Classify](#) models pretrained on the [ImageNet](#) dataset. [Track](#) mode is available for all Detect, Segment and Pose models.



All [Models](#) download automatically from the latest Ultralytics [release](#) on first use.



Lab 3.2

Google AI Studio - Robotics Spatial Understanding

Reset session
Reveal on hover

API REQUEST

```
{
  "model": "gemini-robotics-v1.0-preview",
  "contents": {
    "parts": [
      {
        "inlineData": {
          "data": "BASE64_IMAGE_DATA_REDACTED",
          "mimeType": "image/png"
        }
      }
    ],
    "text": "Task: Detect items. Return a JSON array of objects. Each object must have: {"box_2d": [xmin, ymin, xmax, ymax] (coordinates 0-1000)}."
  }
}
```

API RESPONSE

```
{
  "box_2d": [
    193,
    246,
    453,
    322
  ],
  "label": "bag of beans"
},
{
  "box_2d": [
    213,
    354,
    324,
    451
  ]
}
```

GIVE ME: gemin-robotics-v1.0-preview

☐ Enable thinking
Thinking improves the capabilities of the model to reason through tasks, but may produce less desirable results for simple locating tasks. For simple tasks, disable thinking for improved speed and likely better results.

PROMPT
Detect items:

items

Send

Response time: 5.10s

2D bounding boxes

Points

[Upload an image](#)

temperature: 0.5

Activate Windows
Go to Settings to activate Windows.

Lab 3.3

Google AI Studio - MediaPipe

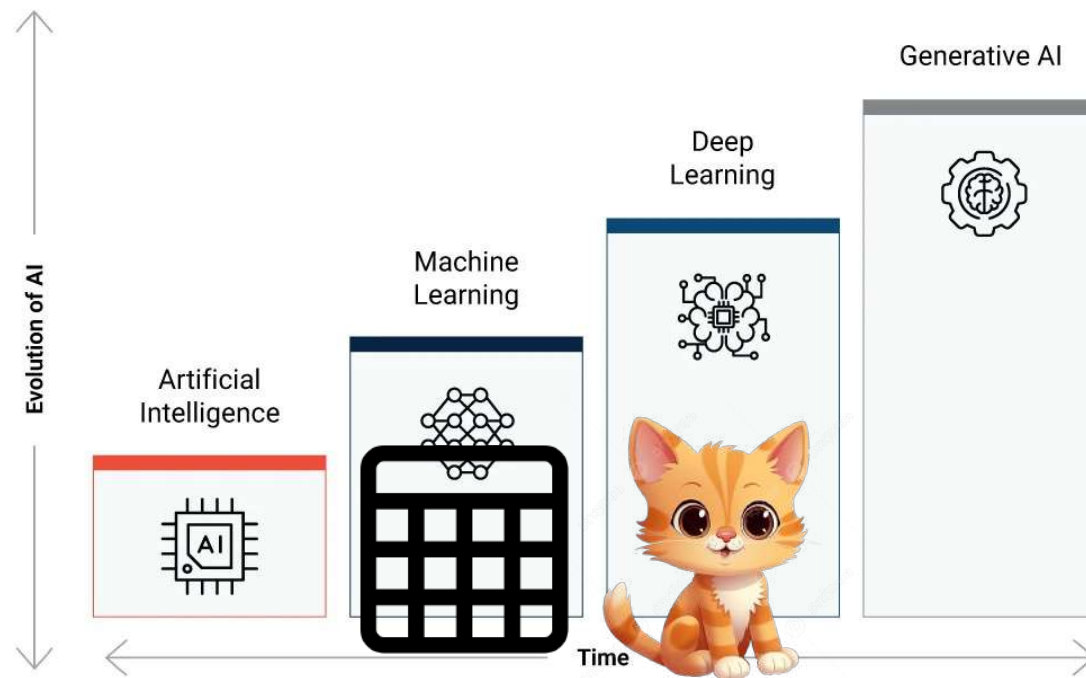
The screenshot displays the Google AI Studio interface, specifically the MediaPipe section. At the top, there is a search bar and a navigation menu with categories: **anana**, **Games and Visualizations**, **GenMedia**, **Multimodal understanding**, **Tools and MCP**, **MediaPipe** (selected), **Code gen**, and **Developer quickstarts**.

The main content area features a large banner for MediaPipe with the text: "Meet MediaPipe: on-device ML solutions! Prompt AI Studio to integrate MediaPipe cross-platform optimized ML solutions automatically." Below this is a "Learn more" button.

Below the banner, there are three featured applications:

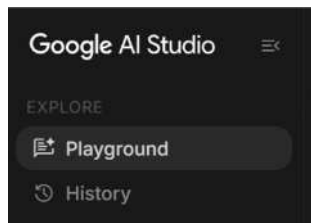
- Vision Sync**: A video feed of a person's face with a "Sadness" emotion label. It uses MediaPipe, Lyria RealTime, and AI Studio. Description: "Uses computer vision to recognize objects and makes music based on what it sees."
- Gemini Slingshot**: A game interface showing a person's face with a slingshot and colorful bubbles. It uses Gemini 3 Flash, Gaming, and MediaPipe. Description: "Gemini 3 Flash is your strategic co-pilot, analyzing the board and suggesting the best shots in this webcam bubble shooter."
- Dino Jump!**: A game interface showing a dinosaur character. It uses MediaPipe, Gaming, and AI Studio. Description: "Burn calories, not just time! Experience the Chrome-Dino-like game powered by MediaPipe - jump in real life to dodge cacti!"

4) Audio → LISTEN / SPEAK

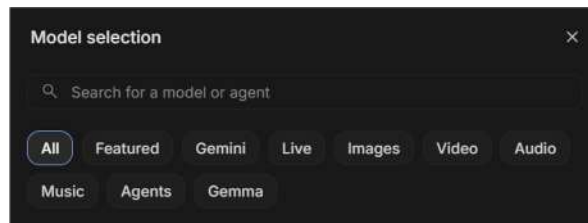


Multimodal Model selection

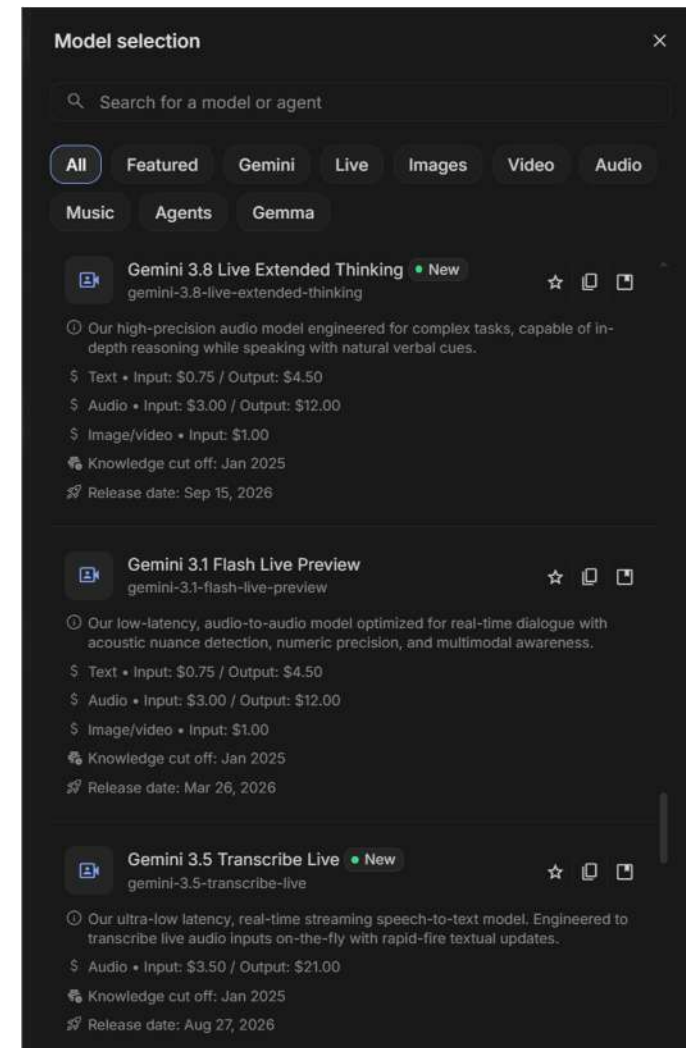
- Multimodal AI อยู่ใน Playground
 - Audio
 - Images
 - Video
 - Agents (Build App)
 - Music



1

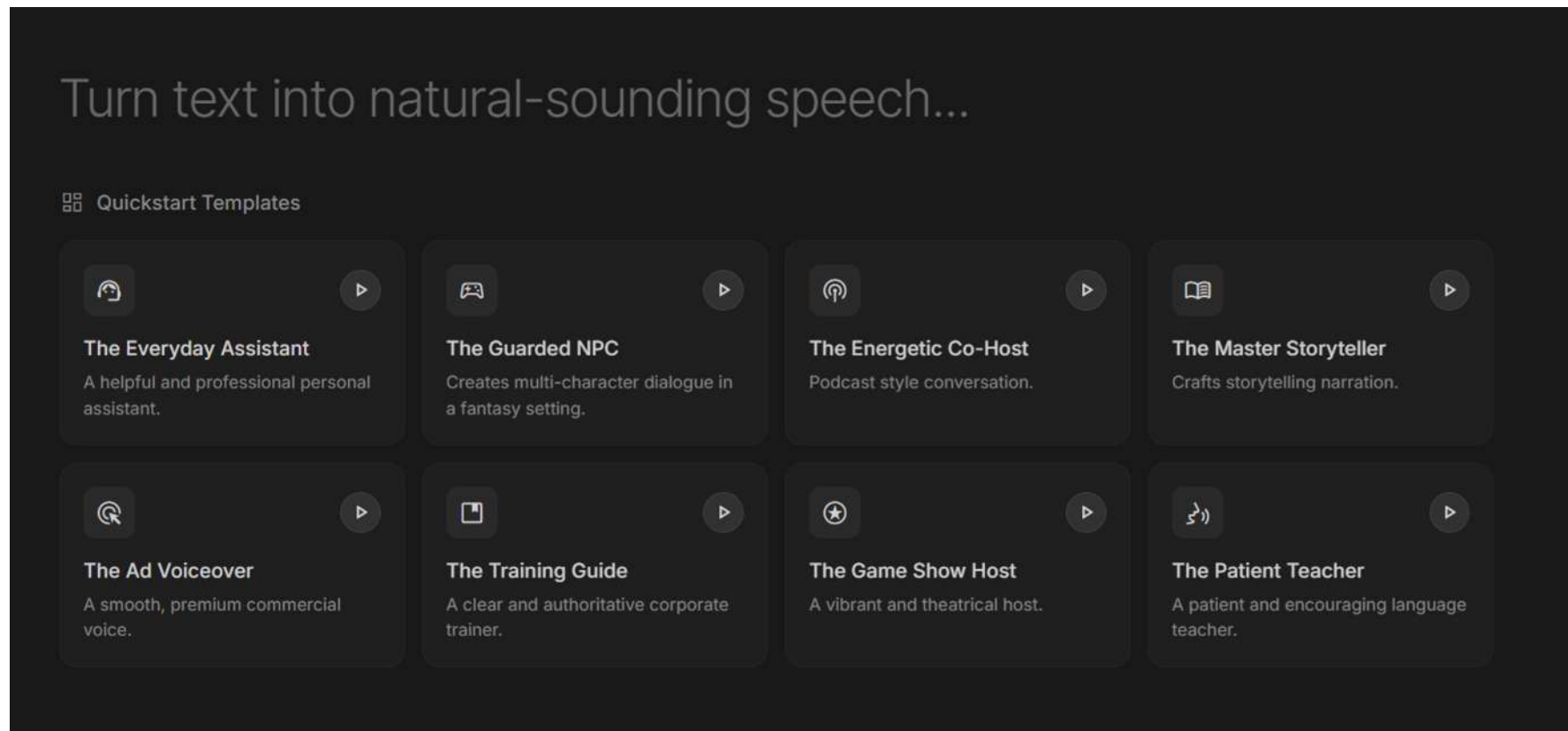


2



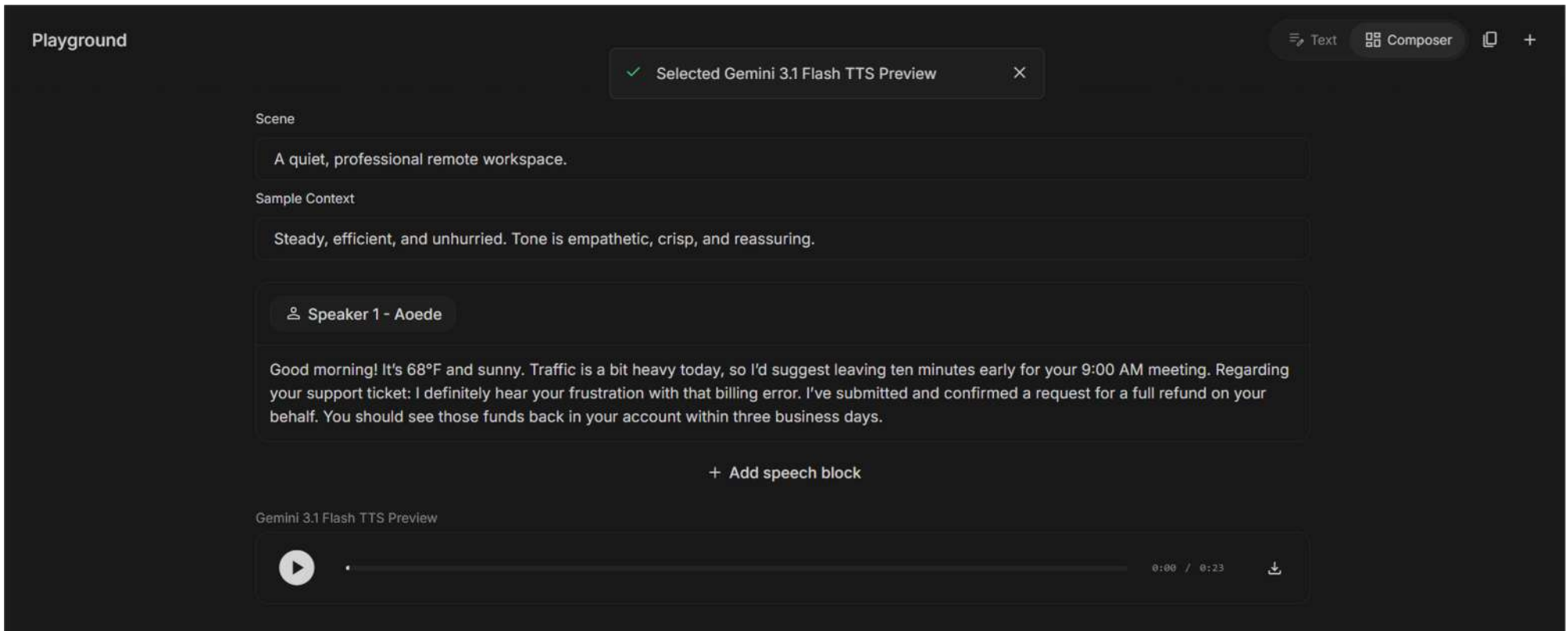
3

Audio (Text-to-Speech)



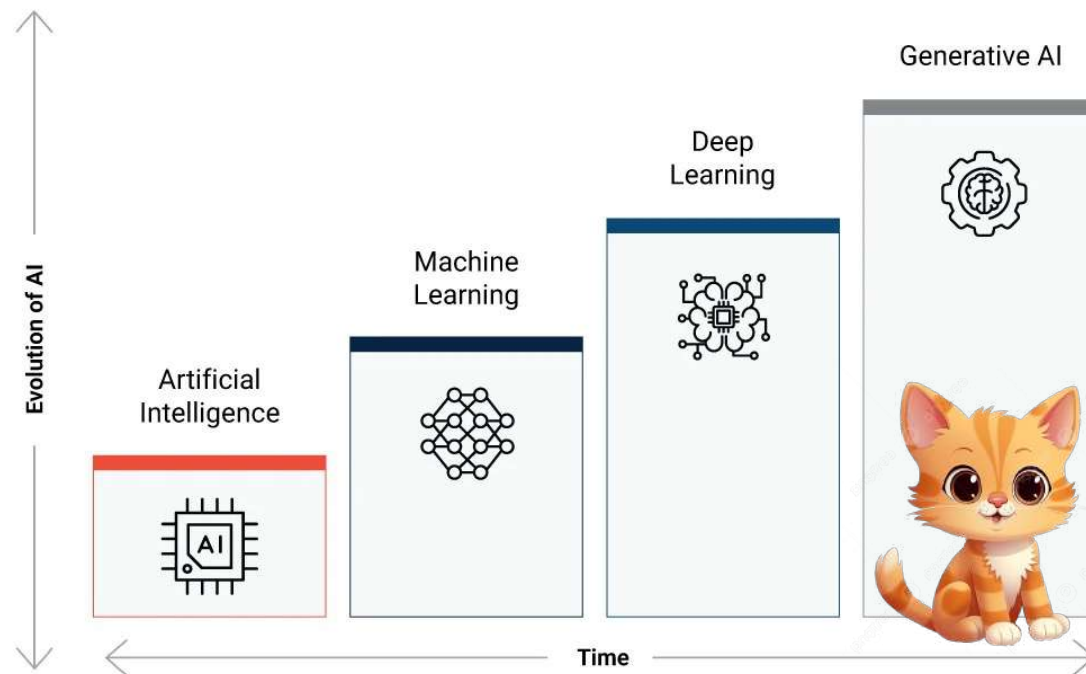
Lab 4

Audio (Text-to-Speech)



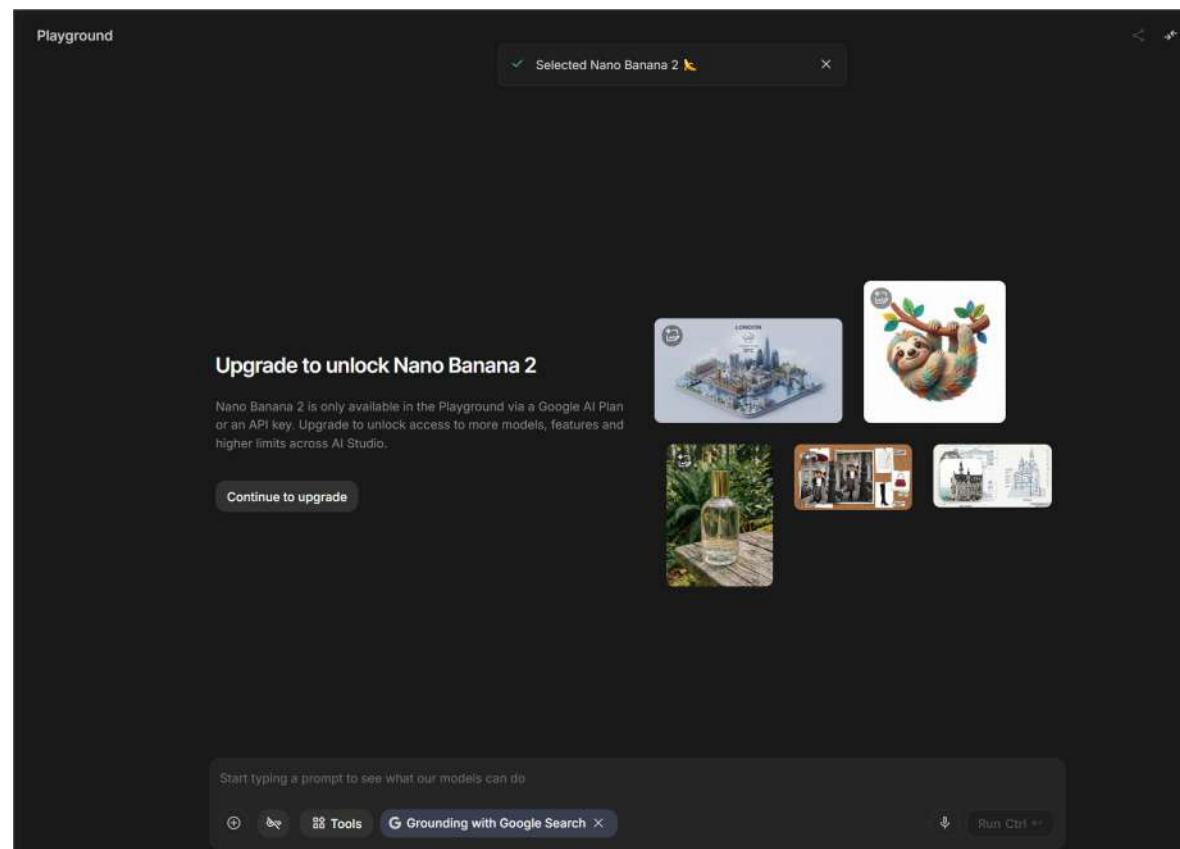
The screenshot displays the Gemini 3.1 Flash TTS Playground interface. At the top left, the word "Playground" is visible. In the top right corner, there are tabs for "Text" and "Composer", along with a copy icon and a plus sign. A notification banner at the top center reads "Selected Gemini 3.1 Flash TTS Preview" with a green checkmark and a close button. The interface is divided into several sections: "Scene" with the text "A quiet, professional remote workspace.", "Sample Context" with the text "Steady, efficient, and unhurried. Tone is empathetic, crisp, and reassuring.", and a speaker selection dropdown currently set to "Speaker 1 - Aoede". Below the speaker selection is a text input area containing a sample script: "Good morning! It's 68°F and sunny. Traffic is a bit heavy today, so I'd suggest leaving ten minutes early for your 9:00 AM meeting. Regarding your support ticket: I definitely hear your frustration with that billing error. I've submitted and confirmed a request for a full refund on your behalf. You should see those funds back in your account within three business days." Below the text input is a button labeled "+ Add speech block". At the bottom, there is a "Gemini 3.1 Flash TTS Preview" section featuring a play button, a progress bar, and a download icon. The progress bar shows a duration of 0:00 / 0:23.

5) ✨ GenAI → GENERATE

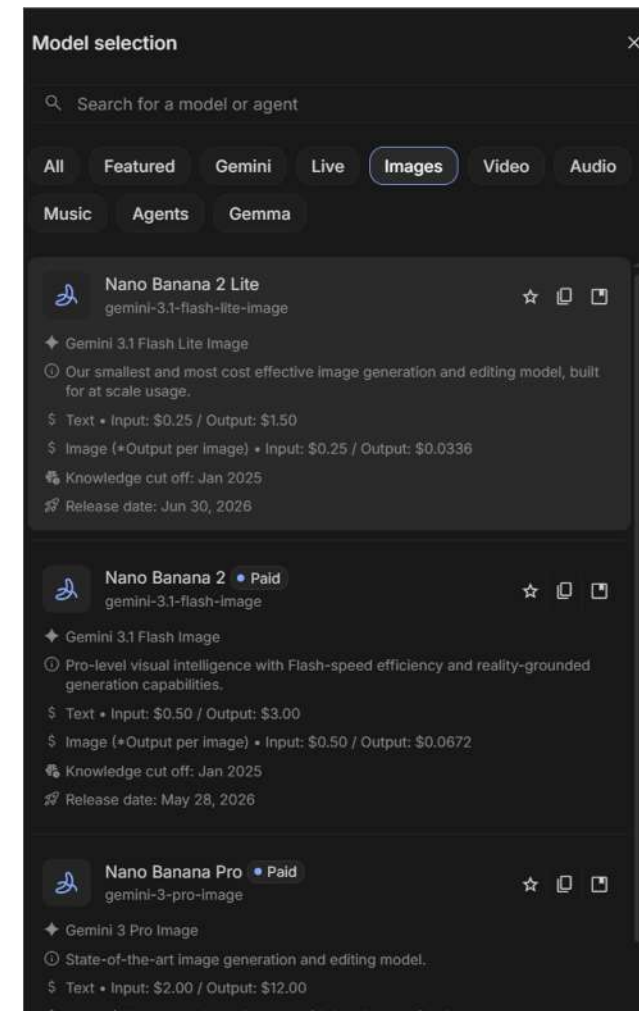
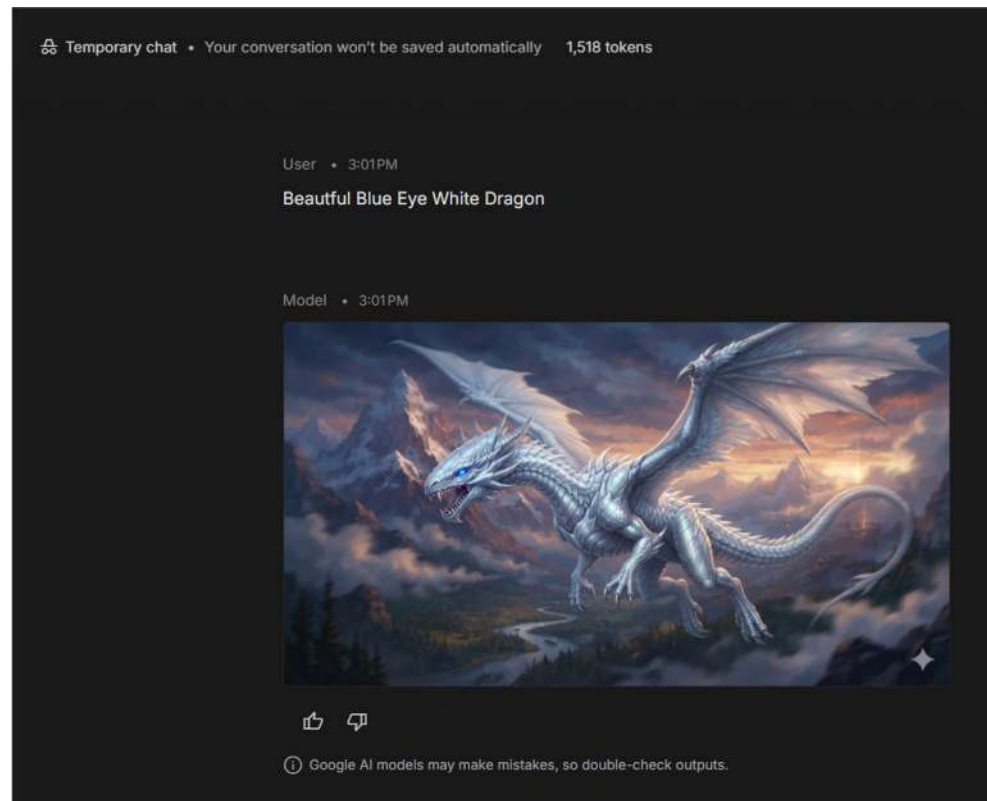


Lab 5.1

Images

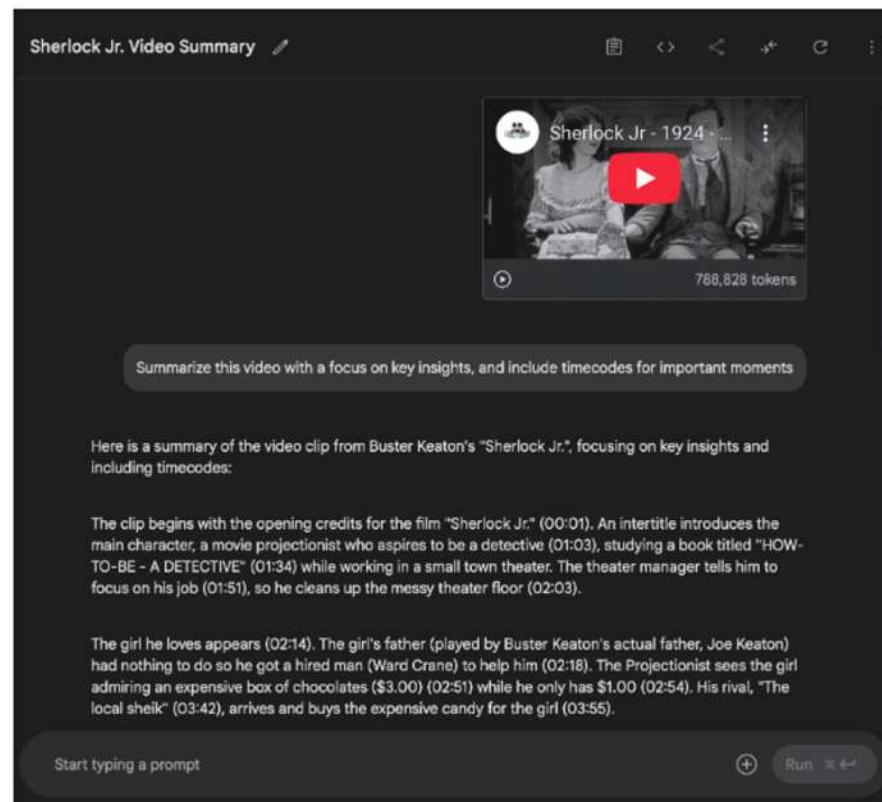


Images



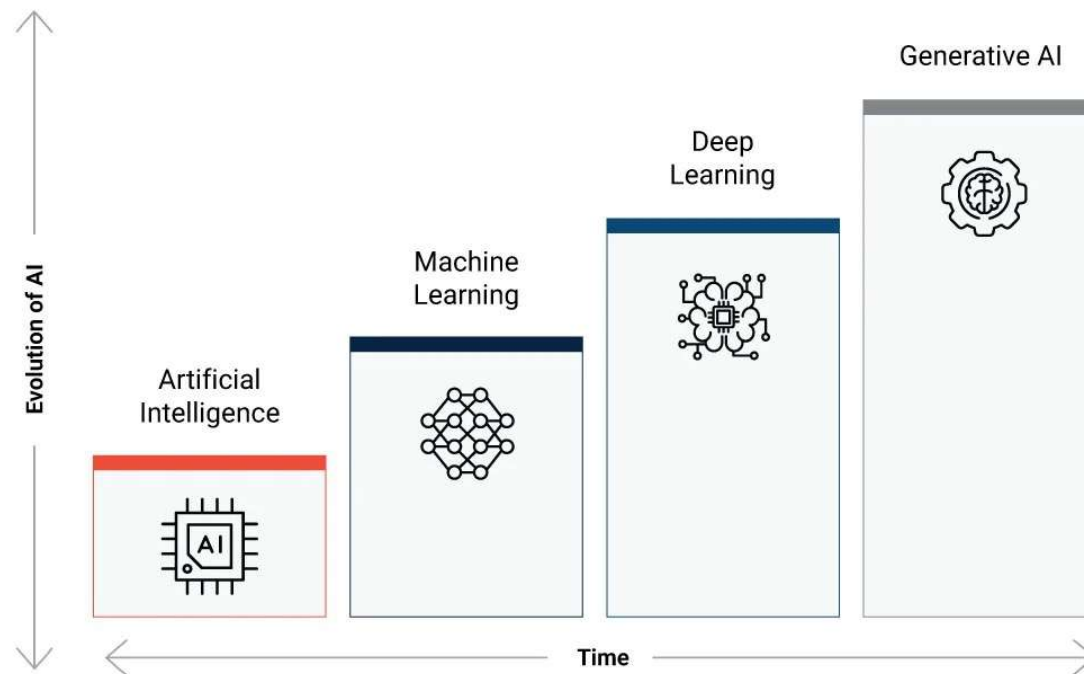
Lab 5.2: Gemini 3.8 Flash

Google AI Studio – Video Summarization



ใส่วิดีโอเข้าไป แล้ว AI สรุปเนื้อหาให้ได้ในไม่กี่วินาที

6) AI Coding → BUILD



Lab 6

Google AI Studio - Gallery

The screenshot displays the Google AI Studio Gallery interface. On the left is a dark sidebar with navigation options: 'EXPLORE' (Playground, History), 'BUILD' (New app, My apps, Gallery), and 'MANAGE' (Dashboard, Documentation). The main area features a search bar at the top and a row of category buttons: 'Featured', 'All apps', 'Gemini 3', 'Nano Banana', 'Games and Visualizations', 'GenMedia', 'Multimodal understanding', and 'Tools and MCP'. A large banner for 'Gemini 3.8 Flash is here' is prominent, followed by a 'Start building' button. Below this, the 'Create with Gemini Flash' section showcases two featured apps: 'The Urban Hive Project' (an interactive annual report) and 'Dinosaur Fossil Map Explorer' (a technical field survey visualization). At the bottom right, there is a 'Windows' watermark and an 'Activate Windows' message.

Google AI Studio

Search across hundreds of apps created using Gemini models on AI Studio...

EXPLORE

- Playground
- History

BUILD

- New app
- My apps
- Gallery

MANAGE

- Dashboard
- Documentation

Featured

All apps

Gemini 3

Nano Banana

Games and Visualizations

GenMedia

Multimodal understanding

Tools and MCP

Gemini 3.8 Flash is here

Gemini 3.8 Flash just launched, bringing noticeable improvements to both output quality and the overall builder experience in Google AI Studio.

Start building

Create with Gemini Flash

The Urban Hive Project

Interactive Annual Report

Gemini 3.7 Flash Interactive Data Visualization

Dinosaur Fossil Map Explorer

Games and Visualizations Nano Banana 2 Lite Gemini 3.8 Flash

Upgrade to unlock more

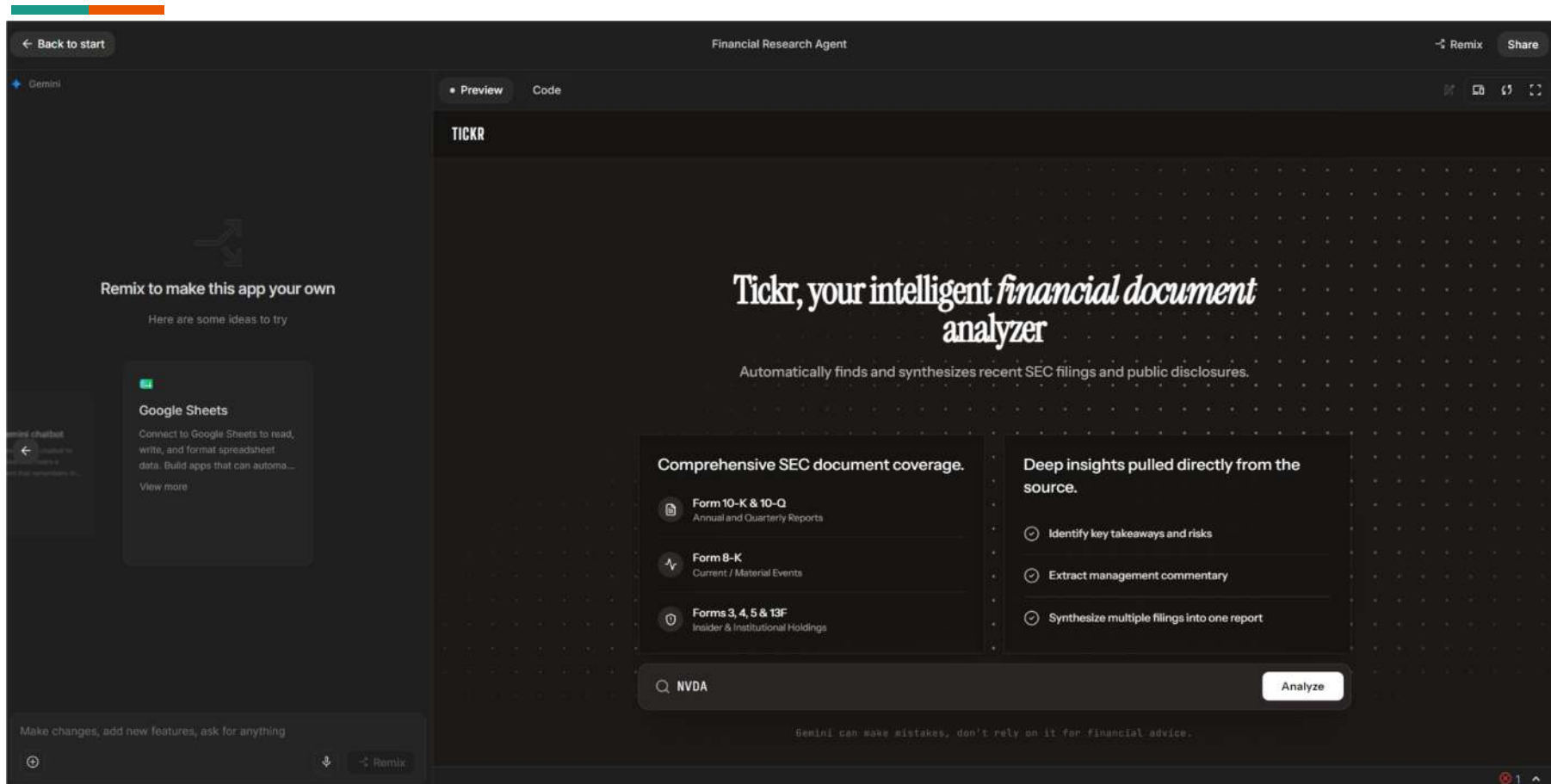
Access higher limits, Pro models, and more.

6772077021@student.ch...

Activate Windows

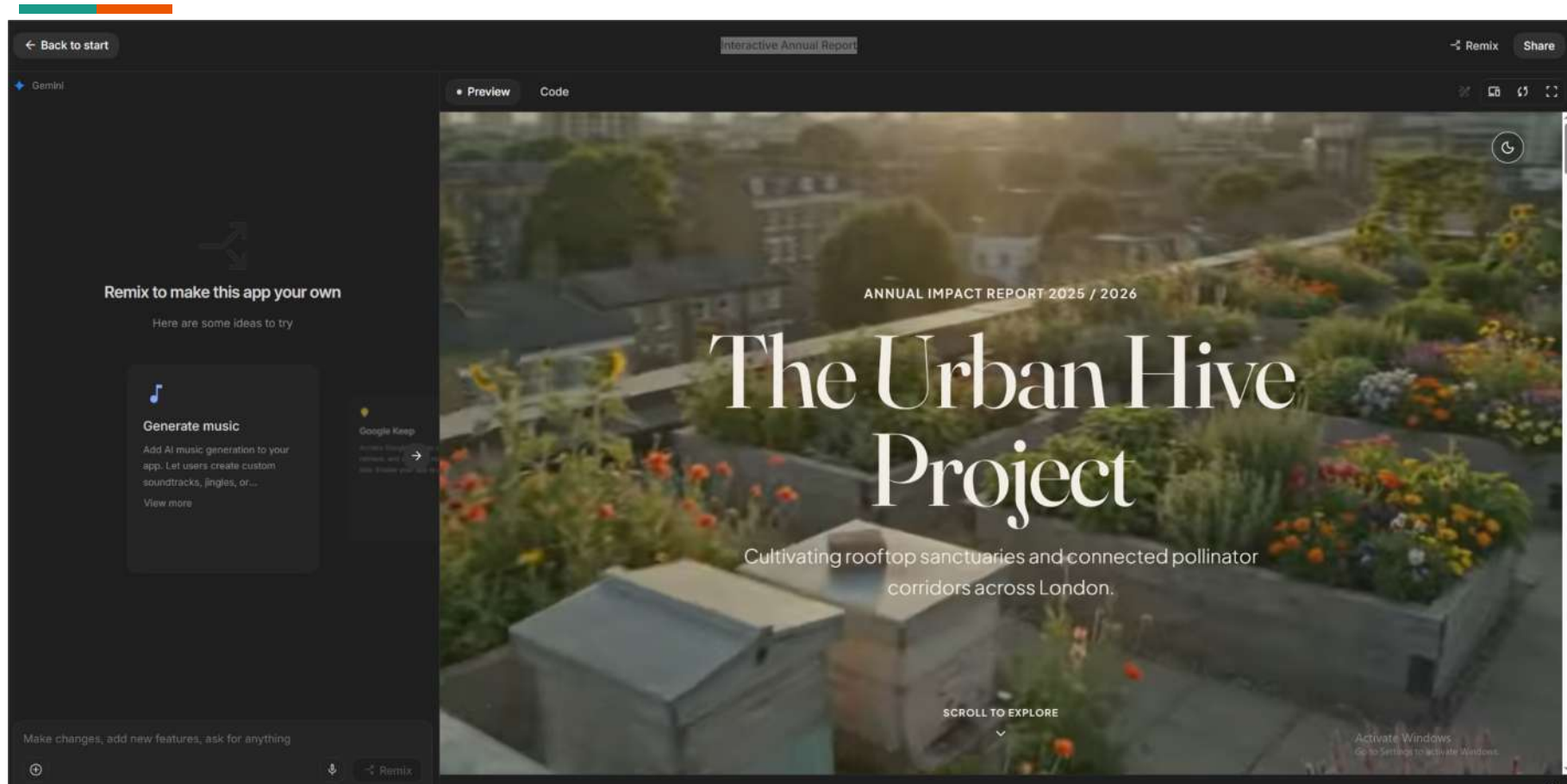
Go to Settings to activate Windows.

Google AI Studio - Financial Research Agent



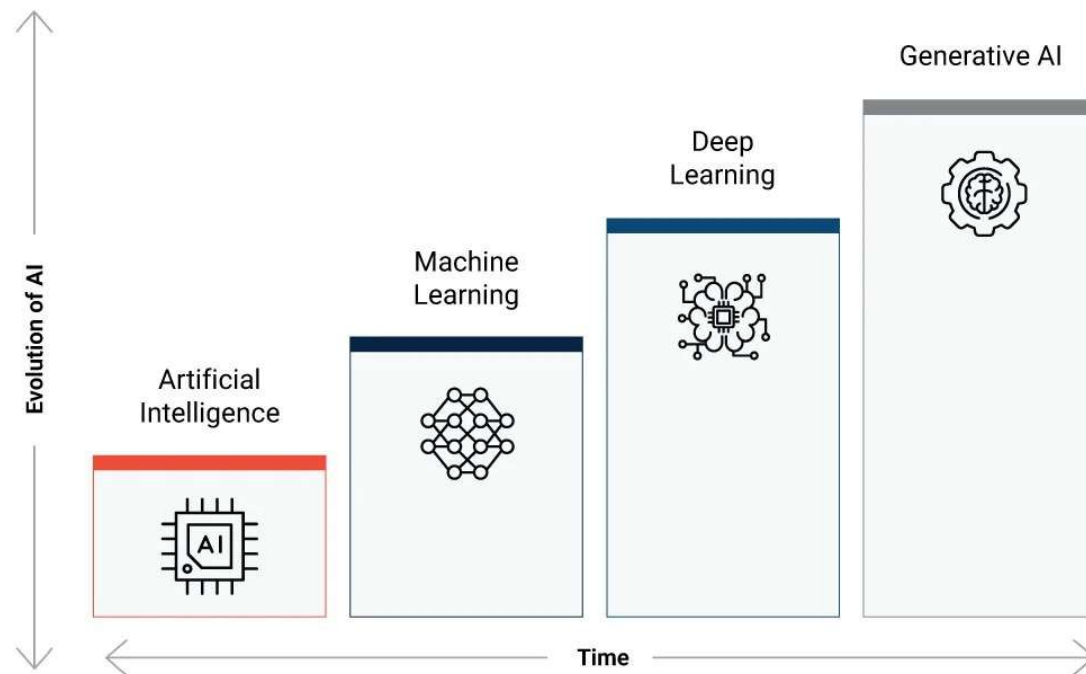
[Link](#)

Google AI Studio - Interactive Annual Report



[Link](#)

7) 🤖 Agent → ACT



Lab 7

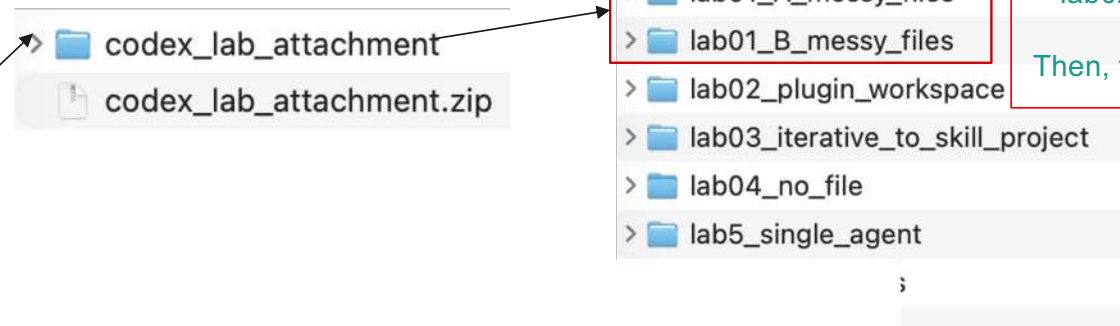
What Is OpenAI Codex?

- Codex is an AI coding and automation agent that can **inspect project files, plan tasks, edit files, run commands, use tools, and produce working outputs under user's environments.**



Lab 1: Handle Local Files + Plan Mode (aka. Lab 1A)

Load prepared files:



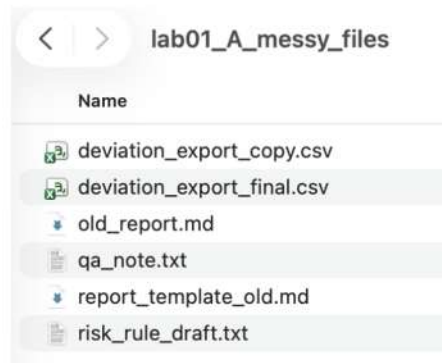
Choose folder

- lab01_A_messy_files for Lab 1A
- lab02_A_messy_files for Lab 1A

Then, follow the steps in lab sheet.

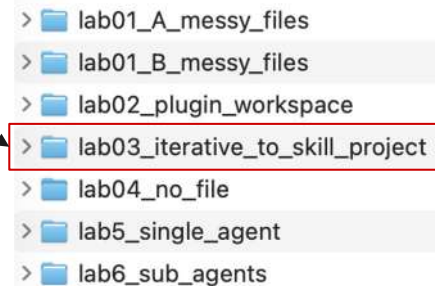
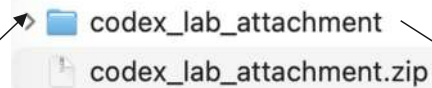
Task:

You are asked to handle a messy file directory containing duplicate files, backup files, temporary files, and old files.



Lab 2: Create Your Own Reusable Skills (aka. Lab 3)

Load prepared files:



Analyze data/raw/food_deviation_log.csv and create a QA report as a Word document using [Documents](#)

Create:
reports/risk_report.docx

Focus on the main food safety and quality risks.
Do not modify files under data/raw/.
Do not use LibreOffice for validation; keep the final DOCX and explain the limitation briefly.

3:38 PM

Lab 2: Create Your Own Reusable Skills

A	B	C	D	E	F	G	H	I	J	K	L
date	product	process_step	issue_type	severity	root_cause	action_taken	downtime_m	batch_id	detected_by	facility	shift
13/5/2026	Ready-to-eat	Washing	Chlorine leve	High	Sanitizer dos	Adjusted dos	32	B-2001	QA Inspector	Plant A	Morning
13/5/2026	Frozen Dumg	Cooking	Core temper	Critical	Steam press	Extended coc	88	B-2002	QA Inspector	Plant A	Evening
14/5/2026	Cold-presset	Filling	Underfilled b	Medium	Filler head n	Cleaned fille	24	B-2003	Line Leader	Plant A	Morning
14/5/2026	Fresh-cut Me	Cutting	Size inconsis	Low	New blade g	Adjusted blac	12	B-2004	Production S	Plant B	Evening
15/5/2026	Ready-to-eat	Storage	Temperature	Critical	Cold room de	Moved pallet	82	B-2005	QA Manager	Plant A	Night
15/5/2026	Frozen Dumg	Mixing	Incorrect ing	Critical	Scale calibra	Stopped batc	95	B-2006	Production S	Plant C	Morning
15/5/2026	Frozen Dumg	Mixing	Incorrect ing	Critical	Scale calibra	Stopped batc	95	B-2006	Production S	Plant C	Morning
16/5/2026	Fresh-cut	Rollback	Foreign mater	High	Recall ratio	Released	45	B-2007	QA Inspector	Plant B	Morning

☞ Ran 8 commands

I've got enough analysis to draft the report. I'm moving to document creation now, and I'll keep the structure tight: executive summary, key risk themes, detailed observations, and immediate actions, with a small note on log-quality issues because they matter for QA credibility.

☞ Explored 2 files, ran 3 commands

I've got the risk story now: this log is dominated by cold-chain failures, formulation control problems, and packaging/traceability issues. I'm turning that into a report with a small metric strip and a ranked risk table so the biggest food-safety impacts are obvious at a glance.

I'm writing the generator now. It will read the CSV, normalize the messy log entries for analysis, and emit the finished DOCX into `reports/risk_report.docx` with tables sized explicitly so the report stays readable.

🔗 Creating `build_risk_report.py` +273 -0



risk_report.docx
Document · DOCX

Food Deviation QA Risk Report

Focused review of food safety and quality deviations recorded in May 2026

Raw rows	Unique incidents	High/Critical incidents	Unique downtime
24	23	13	944 min

Source: data/raw/food_deviation_log.csv

Scope note: The analysis below deduplicates one exact duplicate record and normalizes inconsistent severity labels for reporting.

Executive Summary

The log contains 23 unique incidents (from 24 raw rows) and 13 high-or-critical events. Cold-chain control is the dominant risk theme, accounting for 5 incidents and 330 minutes of downtime.

Plant A has the highest incident volume, while Plant C concentrates the most severe events because of formulation, thermal-processing, and contamination failures in frozen dumpling production.

Packaging and traceability controls are also recurring weak points, with label mismatch, seal defects, and cap issues creating avoidable release and recall risk.

Top Risk Themes

Table 1. Ranked risk themes based on unique incidents, severity, and downtime.

Risk theme	Evidence	Why it matters	Immediate focus
Cold-chain control	5 incidents / 330 min	Repeated temperature abuse in storage and transfer raises pathogen growth and shelf-life risk.	Tighten transfer checks, alarm response, and cold-room verification.
Recipe / formulation control	2 incidents / 155 min	Recipe version drift and missing ingredients can change product composition and regulatory status.	Lock version control and require QA sign-off before mixing.
Contamination control	3 incidents / 123 min	Foreign material and metal detector rejection create direct physical contamination and recall exposure.	Strengthen raw screening, equipment inspection, and challenge checks.
Packaging / traceability	3 incidents / 97 min	Label mismatch and line-clearance failures can create mislabeling and traceability errors.	Enforce pre-start label verification and line clearance.
Thermal processing	1 incident / 85 min	Low core temperature indicates an underprocessed product that may not be safe to release.	Review cook validation and steam-pressure monitoring.

Where the Risk Concentrates

Plant A records the most incidents (12) and is dominated by cold-chain, foreign-material, and fill-control issues in salad and juice lines.

Plant B shows mostly packaging and sanitation drift, including size inconsistency, underfill, seal defects, and one chlorine-level excursion.

Plant C has the highest-severity mix: recipe ratio errors, missing ingredient risk, metal-detection rejection, and an under-temperature cook event.

Immediate Actions

Table 2. Priority actions to reduce the highest-probability and highest-severity risks.

Priority	Action	Primary risk addressed
1	Lock cold-room checks at every transfer and require documented alarm response within the shift.	Temperature abuse
2	Make recipe version control and ingredient reconciliation mandatory before mixing starts.	Incorrect ingredient ratio / missing ingredient
3	Strengthen metal-detector checks, equipment wear inspection, and raw-material screening.	Foreign material / metal contamination
4	Require line clearance and label verification before start-up and after every changeover.	Label mismatch / traceability
5	Standardize the log fields: one severity taxonomy, one date format, and mandatory root cause entry.	Reporting quality

Data Quality Notes

Duplicate entry: One exact duplicate row appears for the 2026-05-03 Frozen Dumpling / Incorrect ingredient ratio event; the report analyzes unique incidents to avoid double counting.

Severity coding: Severity values are inconsistent in the source file: one row is blank and one uses lowercase 'critical'.

Date format: The log mixes date styles such as 2026-05-04, 2026/05/04, and 05/05/2026; this should be standardized for downstream QA tracking.

Missing root cause: The 'Missing ingredient' record has an empty root-cause field, which weakens corrective-action follow-up.

Lab 2: Create Your Own Reusable Skills

Read reports/risk_report.docx and data/raw/food_deviation_log.csv.

Create presentation-ready PNG plots under reports/figures/ and insert them in appropriate places in the existing Word report:

1. Downtime minutes by issue type, sorted descending.
2. Deviation count by severity, normalizing severity case where appropriate.

Use readable labels, clean styling, sensible colors, titles, and short captions explaining each key insight.
Do not modify files under data/raw/.
Do not use LibreOffice for validation.

I've confirmed the report structure, so I'm adding the visuals in a way that keeps the chart for downtime concentration and one for severity distribution, each with its own caption saved under reports/figures/.



risk_report.docx
Document · DOCX



downtime_by_issue_type.png
Image · PNG



deviation_count_by_severity.png
Image · PNG

```
build_risk_report.py X
Users > prommin > Downloads > codex_lab_attachment > lab03_iterative_to_skill_project > build_risk_report
1 from __future__ import annotations
2
3 import csv
4 from collections import Counter, defaultdict
5 from datetime import date
6 from pathlib import Path
7
8 import matplotlib.pyplot as plt
9 from docx import Document
10 from docx.enum.section import WD_SECTION
11 from docx.enum.table import WD_TABLE_ALIGNMENT, WD_CELL_VERTICAL_ALIGNMENT
12 from docx.enum.text import WD_ALIGN_PARAGRAPH
13 from docx.oxml import OxmlElement
14 from docx.oxml.ns import qn
15 from docx.shared import Inches, Pt, RGBColor
16
17
18 ROOT = Path(__file__).resolve().parent
19 CSV_PATH = ROOT / "data" / "raw" / "food_deviation_log.csv"
20 OUT_PATH = ROOT / "reports" / "risk_report.docx"
21 FIG_DIR = ROOT / "reports" / "figures"
22
23
24 def set_run_font(run, name="Calibri", size=None, color=None, bold=None, italic=None):
25     run.font.name = name
26     rpr = run._element.get_or_add_rPr()
27     rfonts = rpr.rFonts
28     rfonts.set(qn("w:ascii"), name)
29     rfonts.set(qn("w:ahAscii"), name)
30     if size is not None:
31         run.font.size = Pt(size)
32     if color is not None:
33         if isinstance(color, RGBColor):
34             run.font.color.rgb = color
35         else:
36             run.font.color.rgb = RGBColor.from_string(color.replace("#", ""))
37     if bold is not None:
38         run.bold = bold
39     if italic is not None:
```

risk_report.docx			
1/1			
75%			
Contamination control	3 incidents / 123 min	foreign material and metal detector rejection create direct physical contamination and recall exposure.	strengthen raw screening, equipment inspection, and challenge checks.
Packaging / traceability	3 incidents / 97 min	Label mismatch and line-clearance failures can create mislabeling and traceability errors.	Enforce pre-start label verification and line clearance.
Thermal processing	1 incident / 85 min	Low core temperature indicates an underprocessed product that may not be safe to release.	Review cook validation and steam-pressure monitoring.

Figures

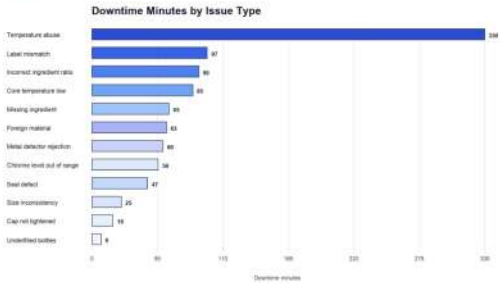


Figure 1. Downtime is concentrated in temperature abuse and recipe control issues, which create the largest operational and food-safety impact.

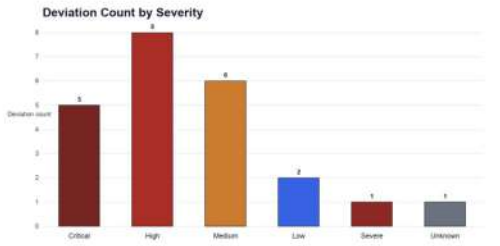


Figure 2. Most deviations are High or Medium severity, but the Critical and Severe events deserve disproportionate attention because they map to the highest food-safety risk.

Where the Risk Concentrates

Plant A records the most incidents (12) and is dominated by cold-chain, foreign-material, and fill-control issues in salad and juice lines.

Plant B shows mostly packaging and sanitation drift, including size inconsistency, underfill, seal defects, and one chlorine-level excursion.

Plant C has the highest-severity mix: recipe ratio errors, missing ingredient risk, metal-detection rejection, and an under-temperature cook event.

Immediate Actions

Table 2. Priority actions to reduce the highest-probability and highest-severity risks.

Lab 2: Create Your Own Reusable Skills

This output is good. Turn this workflow into a reusable Codex skill template named `qa-report-w-visualize`.

Create a reusable Codex skill in the user's installed skill directory so it can be discovered in future sessions.

```
python - <<'PY'  
from pathlib import Path  
base = Path.home() / '.codex' / 'skills' / 'qa-report-w-visualize'...
```

Expand

1. Yes

Click `Yes` so you can call this skill `everywhere` even outside this project directory.

2. Yes, and don't ask again this session

3. No, and tell Codex what to do differently

Skip

Submit

↑ ↓



SKILL.md
Document · MD

Finally, [SKILL.md](#) will be provided as reusable skill for any agents.

SKILL.md +

Users > prominin > .codex > skills > qa-report-w-visualize > SKILL.md

Metadata	
name	qa-report-w-visualize
description	Create food-safety or quality QA report DOCX files from CSV deviation logs, add presentation-ready charts, and place them into an existing or new Word report.

QA Report with Visualizations

Use this skill when the user wants a QA or risk report built from a deviation log, especially when the output should include presentation-ready charts inside a Word document.

Core workflow

1. Read the source CSV and the current DOCX, if one exists.
2. Normalize obvious reporting issues for analysis only, such as duplicate rows, inconsistent severity casing, and mixed date formats. Do not modify the raw data file.
3. Identify the main safety and quality risks using counts, severity, and downtime.
4. Create clean PNG figures under `reports/figures/` with readable labels, titles, and short insight captions.
5. Insert the figures into the report near the relevant analysis sections.
6. Keep the report concise and decision-oriented: executive summary, ranked risks, figures, actions, and a short data-quality note.
7. Verify the final DOCX structurally. Do not use LibreOffice for validation.

Default analysis pattern

When the log has fields like `issue_type`, `severity`, and `downtime_min`:

- Plot downtime by issue type, sorted descending.
- Plot deviation count by severity, normalizing severity case where needed.
- Call out the highest-risk themes first, not the most frequent minor defects.
- Mention data-quality issues only if they affect the trustworthiness of the analysis.

Chart guidance

- Use horizontal bars for long issue names and vertical bars for simple categorical counts.
- Use restrained, high-contrast colors and avoid cluttered legends.
- Label bars directly when possible.
- Keep figure captions short and insight-led.
- Save figures as PNGs in `reports/figures/` with stable filenames.

Report guidance

- Preserve an existing report when present; insert figures into the most relevant section.
- If creating a new report, keep the structure simple: title, summary metrics, executive summary, ranked risks, figures, actions, and data-quality notes.
- Do not edit files under `data/raw/`.
- Prefer concise prose over dense tables unless the task truly needs tabular comparison.

Practical checks

- Confirm the DOCX contains the inserted images.
- Confirm chart captions match the insight shown in the figure.
- If rendering or visual QA is unavailable, state that clearly rather than claiming validation.

Lab 2: Create Your Own Reusable Skills

SKILL.md

Users > prominin > .codes > skills > qa-report-w-visualize > SKILL.md

Metadata

name

qa-report-w-visualize

description

Create food-safety or quality QA report DOCX files from CSV deviation logs, add presentation-ready charts, and place them into an existing or new Word report.

QA Report with Visualizations

Use this skill when the user wants a QA or risk report built from a deviation log, especially when the output should include presentation-ready charts inside a Word document.

Core workflow

1. Read the source CSV and the current DOCX, if one exists.
2. Normalize obvious reporting issues for analysis only, such as duplicate rows, inconsistent severity casing, and mixed date formats. Do not modify the raw data file.
3. Identify the main safety and quality risks using counts, severity, and downtime.
4. Create clean PNG figures under reports/figures/ with readable labels, titles, and short insight captions.
5. Insert the figures into the report near the relevant analysis sections.
6. Keep the report concise and decision-oriented: executive summary, ranked risks, figures, actions, and a short data-quality note.
7. Verify the final DOCX structurally. Do not use LibreOffice for validation.

Default analysis pattern

When the log has fields like `issue_type`, `severity`, and `downtime_min`:

- Plot downtime by issue type, sorted descending.
- Plot deviation count by severity, normalizing severity case where needed.
- Call out the highest-risk themes first, not the most frequent minor defects.
- Mention data-quality issues only if they affect the trustworthiness of the analysis.

Chart guidance

- Use horizontal bars for long issue names and vertical bars for simple categorical counts.
- Use restrained, high-contrast colors and avoid cluttered legends.
- Label bars directly when possible.
- Keep figure captions short and insight-led.
- Save figures as PNGs in reports/figures/ with stable filenames.

Report guidance

- Preserve an existing report when present; insert figures into the most relevant section.
- If creating a new report, keep the structure simple: title, summary metrics, executive summary, ranked risks, figures, actions, and data-quality notes.
- Do not edit files under data/raw/.
- Prefer concise prose over dense tables unless the task truly needs tabular comparison.

Practical checks

- Confirm the DOCX contains the inserted images.
- Confirm chart captions match the insight shown in the figure.
- If rendering or visual QA is unavailable, state that clearly rather than claiming validation.

Chats >

...

Qa Report W Visualize

Create food-safety or quality QA report DOCX files from CSV deviation logs, add pres...

Use @qa-report with the uploaded for_global_test.csv and create a new QA risk report.

+ Default permissions

5.4-Mini Medium

Use created skill to automate your work : Read .csv -> Report .docx -> Visualize

QA Risk Report

Food deviation log analysis for food_deviation_log_v2.csv

Source window: 2026-05-13 to 2026-05-26 | Generated on 2026-06-01

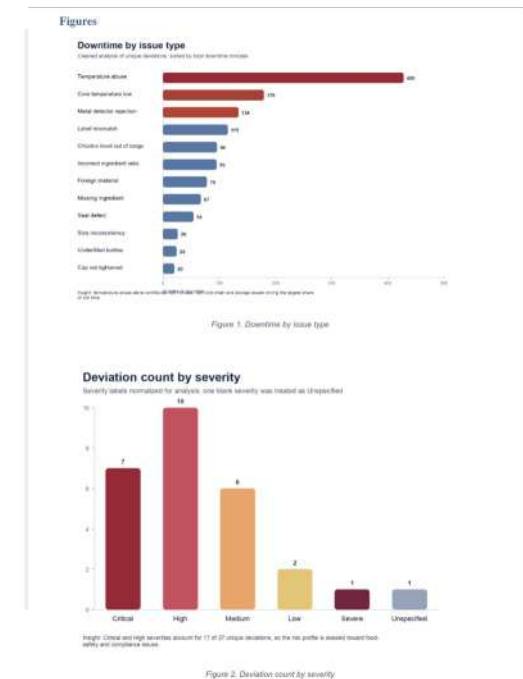
Summary metrics	
Unique deviations	Total downtime
27	1316 min
Critical + High	Facilities impacted
17 records	3

Executive summary

- The log contains 27 unique deviations after removing one exact duplicate row. Critical and High events make up 17 of 27 records (63%), so the risk profile is concentrated in higher-severity issues.
- Temperature abuse is the single largest risk theme at 6 events and 428 minutes of downtime, representing about 33% of all lost time.
- Cold-chain, cooking, and recipe-control failures are the biggest operational drivers, while labeling and metal-detection deviations add traceability and contamination-control risk.

Ranked risks

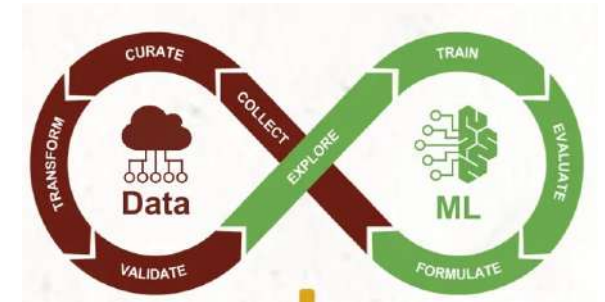
- 1. Temperature abuse - 428 min across 6 event(s); severity mix: Critical 4, Severe 1, High 1.
- 2. Core temperature low - 179 min across 2 event(s); severity mix: Critical 2.
- 3. Metal detector rejection - 134 min across 2 event(s); severity mix: High 2.
- 4. Label mismatch - 115 min across 3 event(s); severity mix: High 2, Medium 1.
- 5. Chlorine level out of range - 96 min across 3 event(s); severity mix: High 2, Unspecified 1.



Digital Dexterity & Data Intelligence:

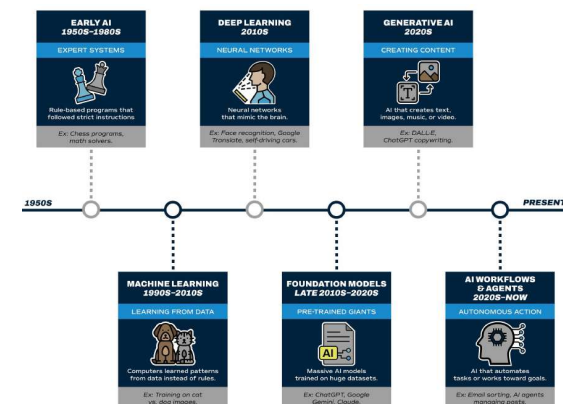
How do we deploy AI models?

Data + ML (recap)



1. 📄 Document → KNOW
2. 📊 Table → PREDICT
3. 📷 Image / Video → SEE
4. 🎤 Audio → LISTEN / SPEAK
5. ✨ GenAI → GENERATE
6. 💻 AI Coding → BUILD
7. 🤖 Agent → ACT

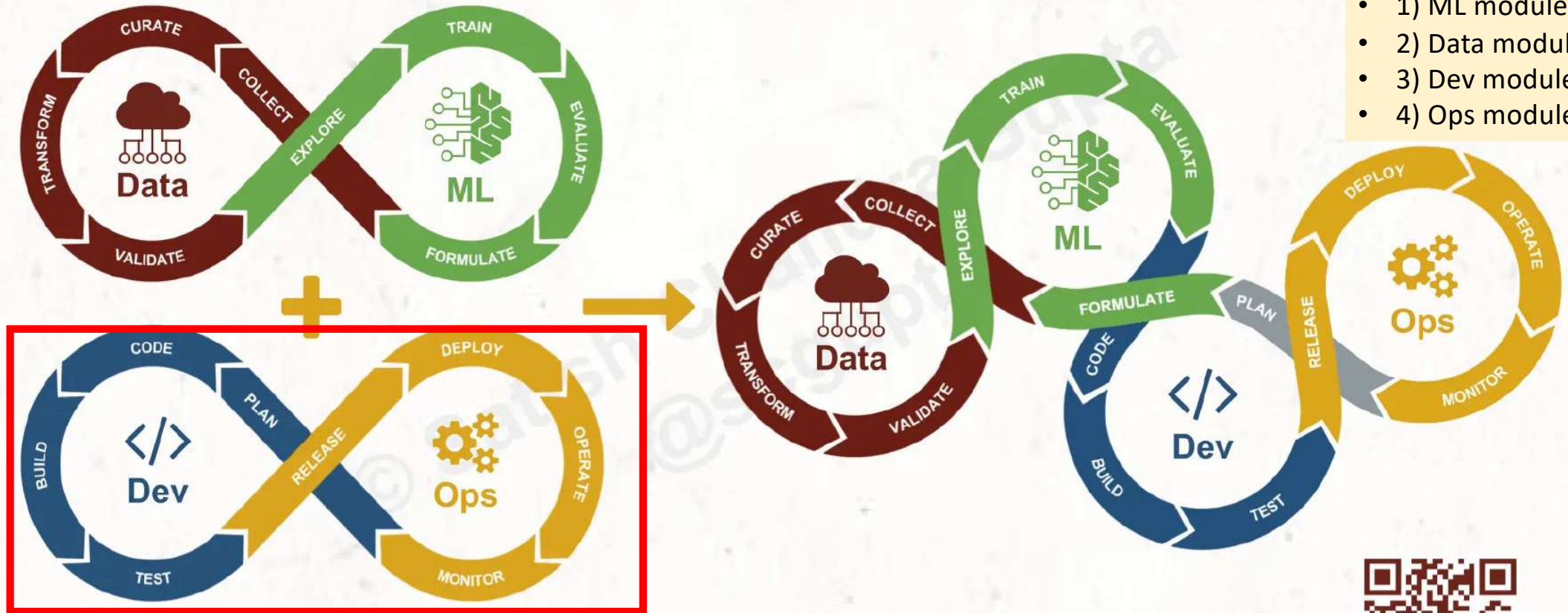
- 💻 1) Software
- 📊 2) Analytics
- 🧠 3) ML/DL
- ✨ 4) GenAI
- 🤖 5) AI Agents



MLOps = DataML + DevOps

ml4devs.com/mlops-lifecycle 

- 1) ML module
- 2) Data module
- 3) Dev module
- 4) Ops module



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<https://www.ml4devs.com/images/illustrations/ml-lifecycle-fusing-model-and-software-development.webp>

scgupta.me 
twitter.com/scgupta 
linkedin.com/in/scgupta 



Do you think that user can use this code to get the prediction result? NO!!!

More than 50% of AI projects were **FAILED** since they didn't plan about the deployment.

2_Linear-Regression-v2.ipynb

File Edit View Insert Runtime Tools Help

+ Code + Text Copy to Drive

```
[ ] 1 coeff_df = pd.DataFrame(lm.coef_, lm.feature_names_in_, columns=['Coefficient'])
    2 coeff_df
```

Does this make sense? Probably not because I made up this data. If you want real data to repeat this sort of analysis, check out the [boston dataset](#).

```
from sklearn.datasets import load_boston
boston = load_boston()
print(boston.DESCR)
boston_df = boston.data
```

▼ Predictions from our Model

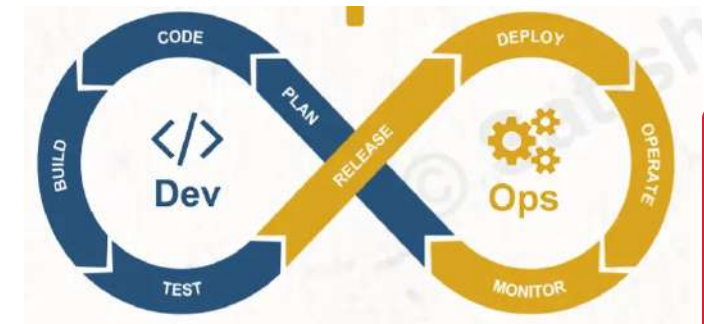
Let's grab predictions off our test set and see how well it did!

```
[ ] 1 predictions = lm.predict(X_test)
```

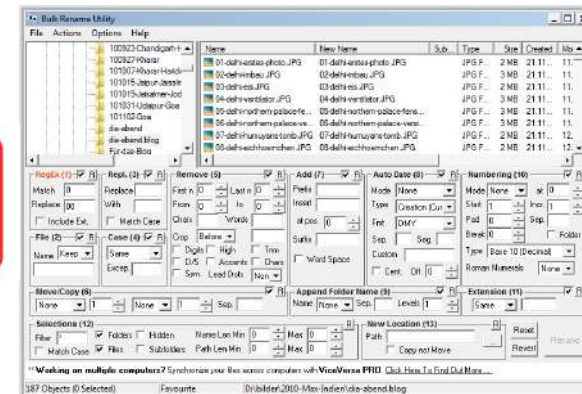
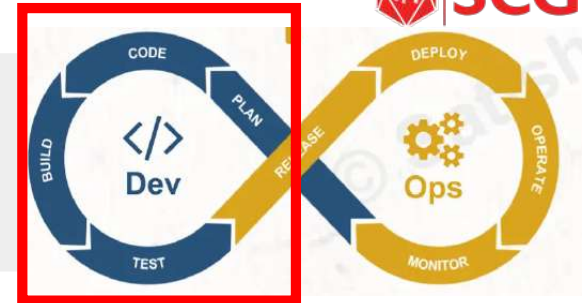
```
[ ] 1 plt.scatter(y_test, predictions)
```

Residual Histogram

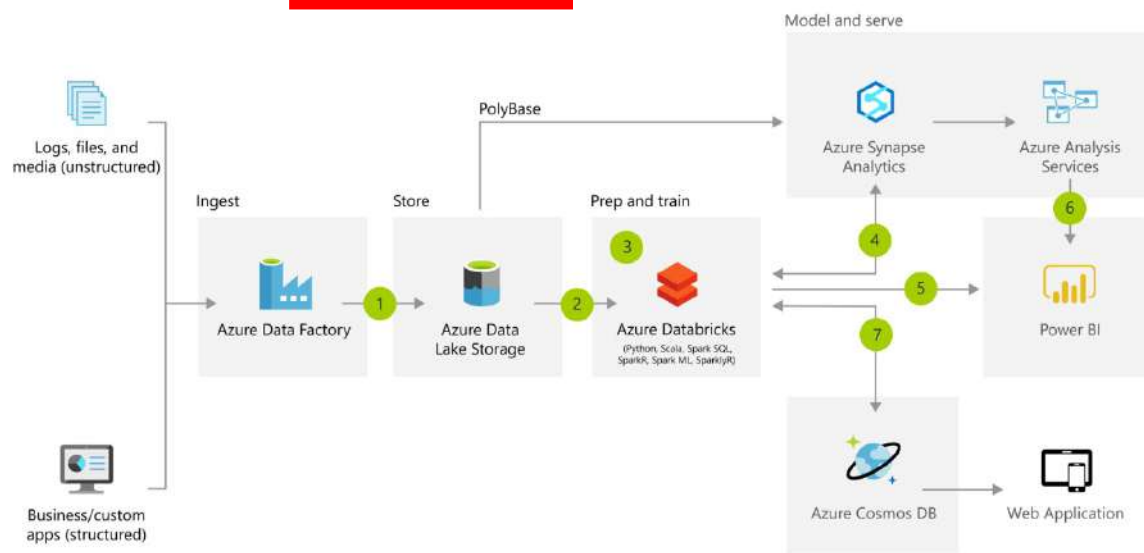
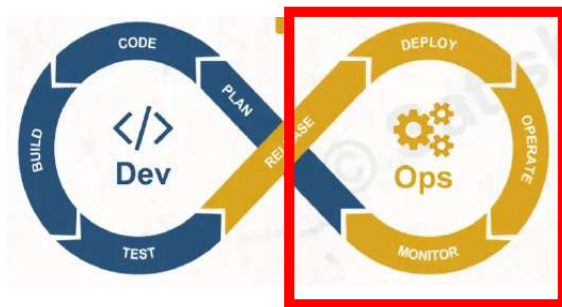
```
[ ] 1 sns.distplot((y_test-predictions), bins=50);
```



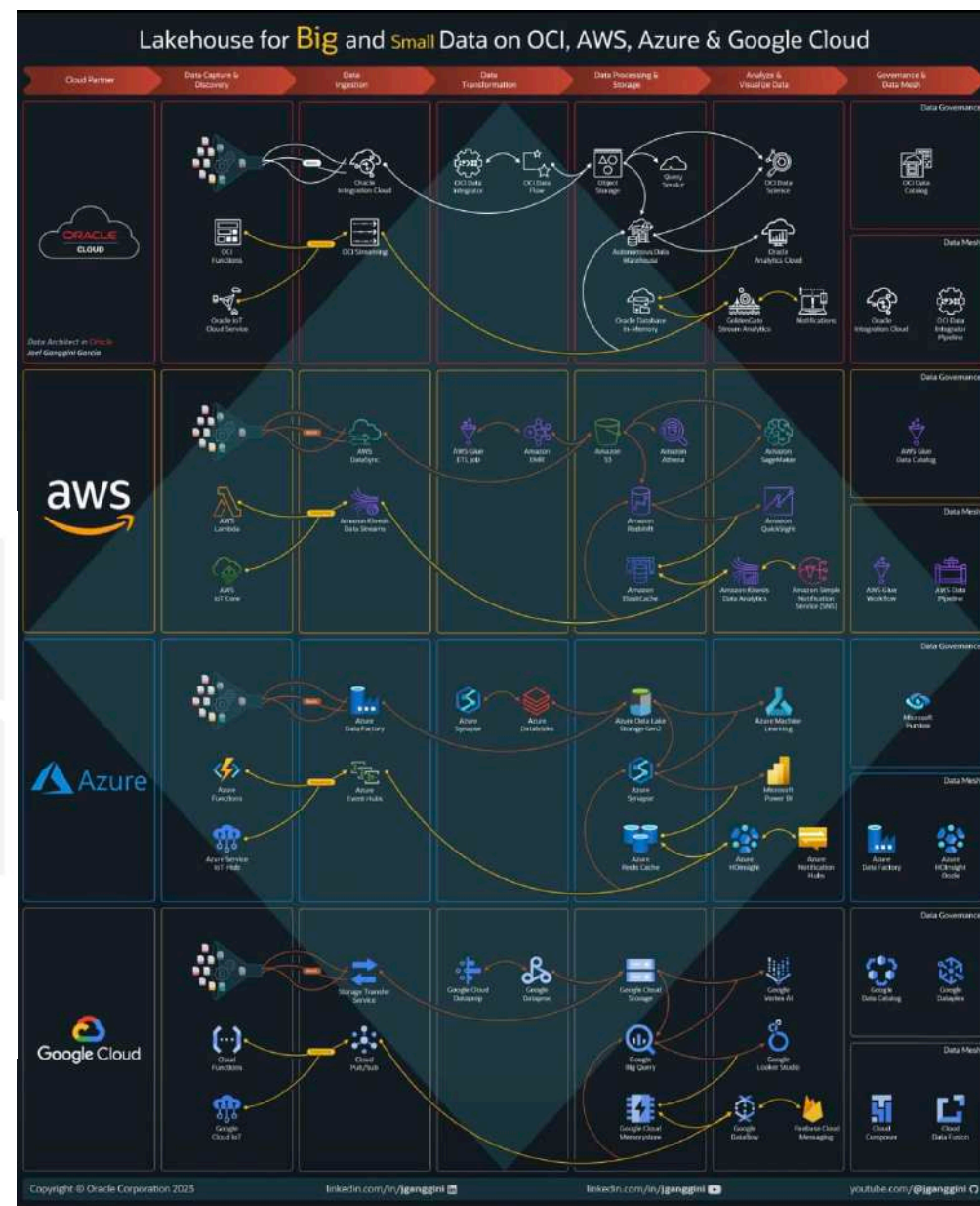
Need to understand touchpoints
Also, need to understand users & journey



Where to deploy?



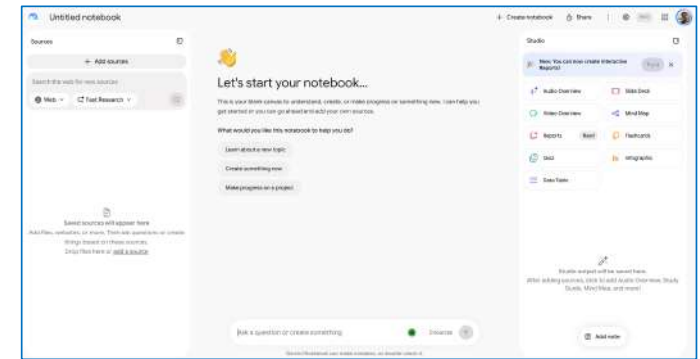
IRIS



Lab 8: Notebook LM

Group Activity

- Example: Targeted Customer Prediction



Section	Example
Problem & Objective	Identify customers most likely to purchase
Data	Customer profile, purchase & campaign history
ML / AI	Predict purchase probability & rank customers
Dev	Sales dashboard with recommended customers
Ops	Update data, predict, monitor & retrain
KPI & Expected Outcome	Conversion ↑, Revenue ↑, Sales effort ↓

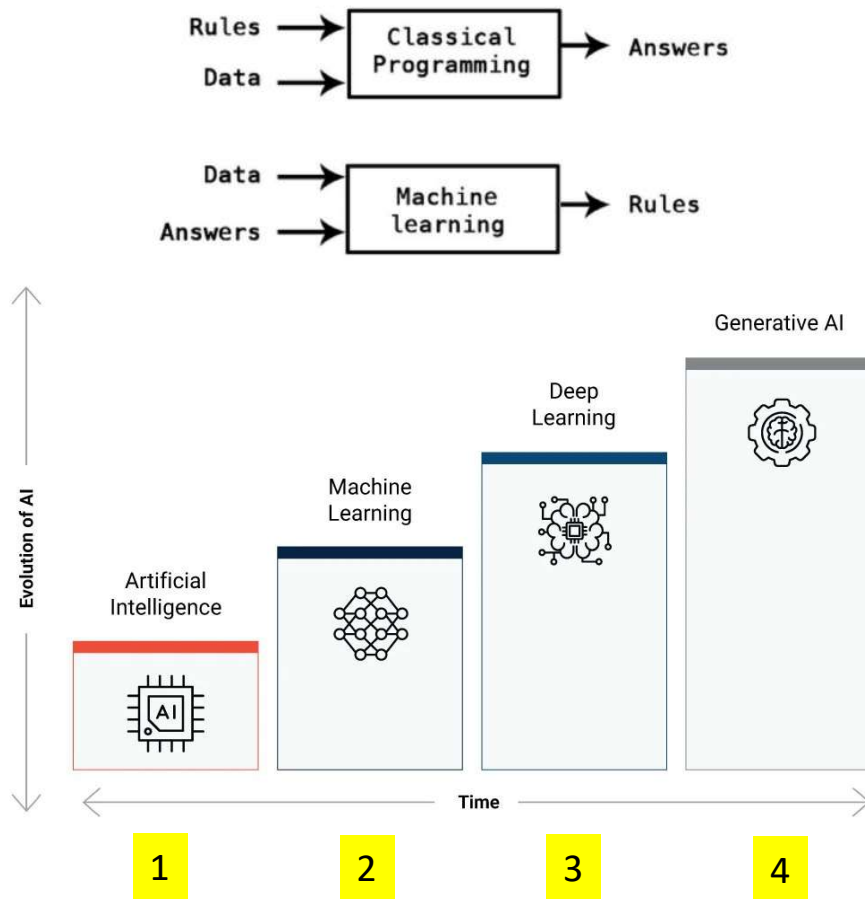
Digital Dexterity & Data Intelligence:

Conclusion

Objective: Digital Dexterity & Data Intelligence

- Be able to understand about recent digital technologies
- Be able to understand the process of applying these technologies
- Be able to understand the data and needs in your department
- Be able to plan and define KPI's for applying these technologies in your department

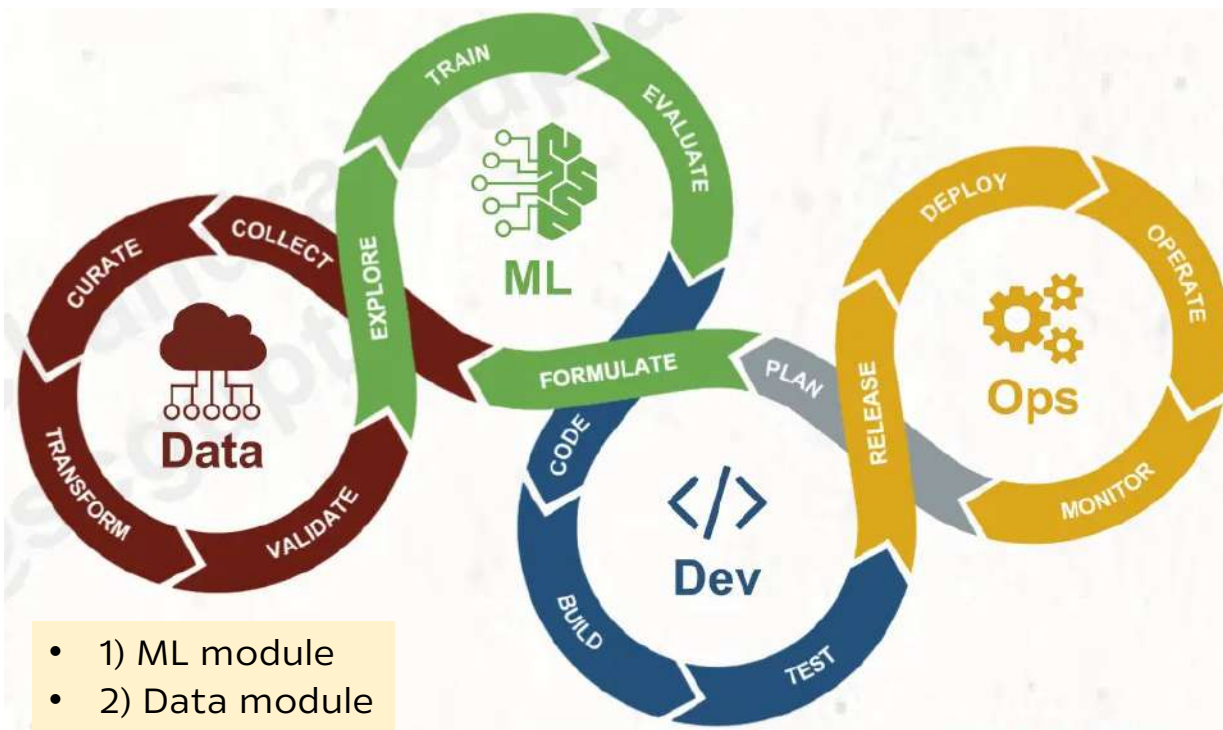
Be able to understand about recent digital technologies (AI)
Be able to understand the data and needs in your department



• Digital & AI Landscape

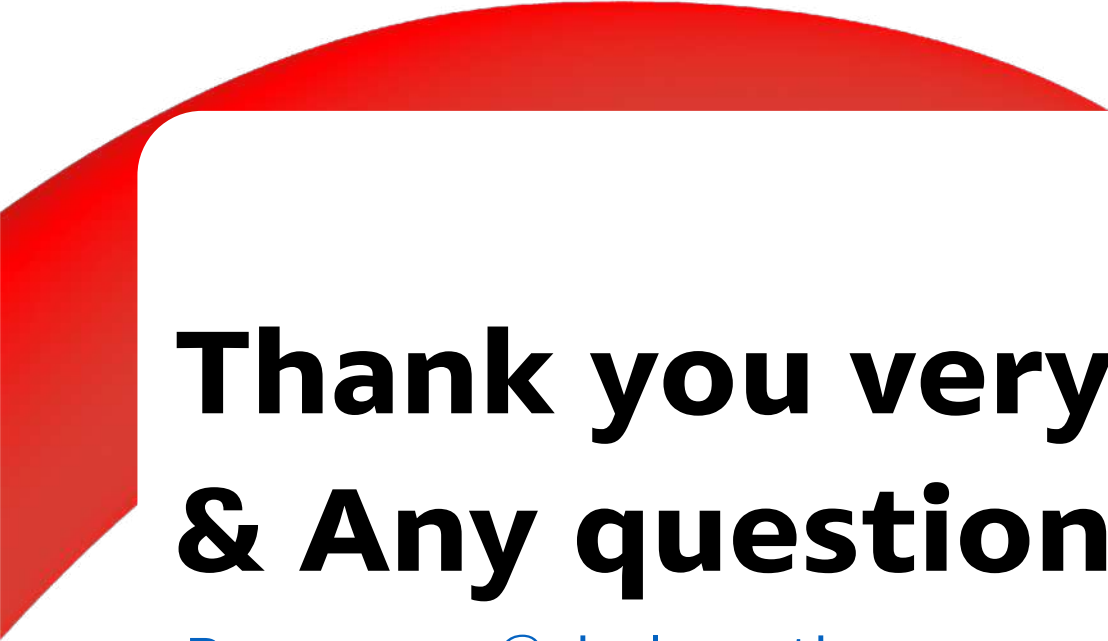
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Be able to understand and use the process of applying these tech
Be able to plan and define KPI's for applying these tech in your department



- 1) ML module
- 2) Data module
- 3) Dev module
- 4) Ops module



A large, stylized red shape on the left side of the slide, resembling a thick, curved arrow pointing right, framing the main text.

Thank you very much & Any questions

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